

SLA OR NAY?



CAUSAL IMPACTS OF BRISTOL DLAS

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WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

SLA OR NAY?

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IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

SLA OR NAY?

DISCRETIONARY LICENSING AUTHORITY

*requirements for privately
rented properties falling
within a council's
jurisdiction that extend
national requirements for
licensing*

DISCRETIONARY LICENSING

DISCRETIONARY
LICENSING
AUTHORITY

ADDITIONAL
LICENSING

*extends national HMO
requirements to “smaller”
households of 3 or more
people w/ shared facilities*

DISCRETIONARY LICENSING

DISCRETIONARY
LICENSING
AUTHORITY

*extends national HMO
requirements to
all properties for 5 years*

ADDITIONAL
LICENSING

SELECTIVE
LICENSING

DISCRETIONARY LICENSING

DISCRETIONARY
LICENSING
AUTHORITY

applies

ADDITIONAL
LICENSING

to certain kinds of properties

SELECTIVE
LICENSING

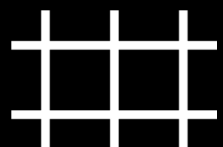
in specific areas & times

DISCRETIONARY LICENSING



EASTERN SLA (1)

Addressing severe quality issues
2016-2021 in St. George West & Eastville
All privately-let properties
EPC, Electric, Gas safety inspections
£300 to £750 fee



CENTRAL ALA (2)

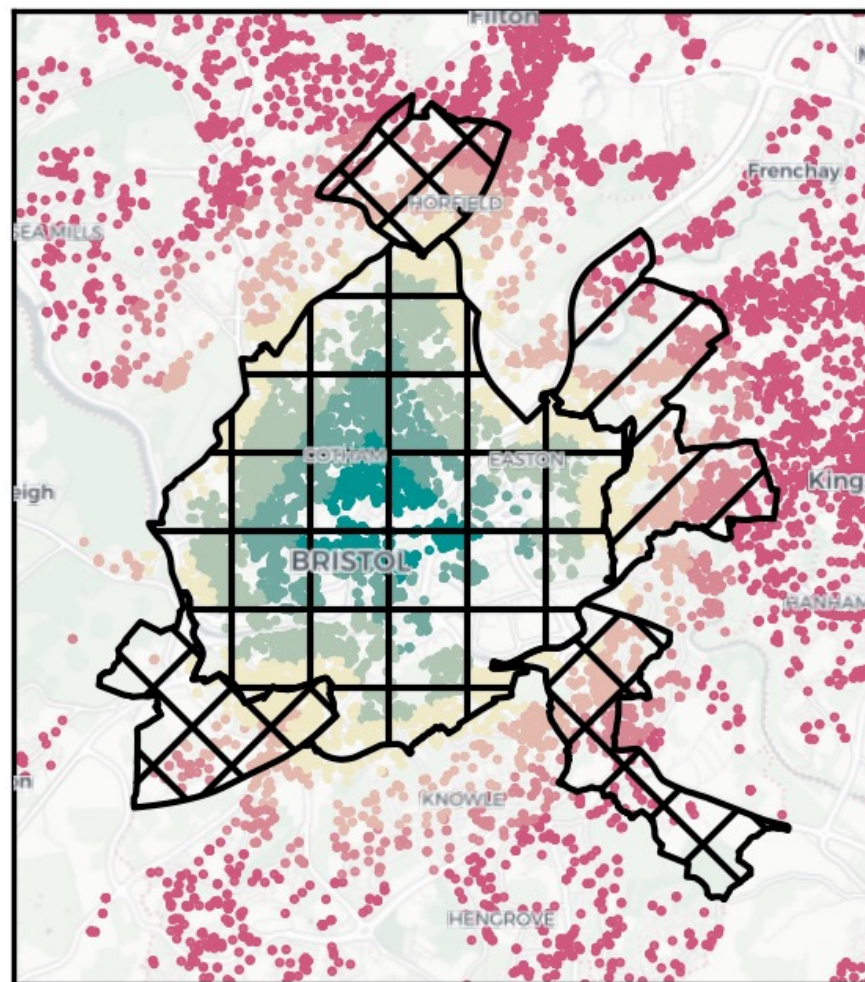
2019-2024 in all central wards
3+ People w/ Shared Facilities
EPC, Electric, Gas safety inspections
£900 to £1450 fee



EXTENSION (3)

2024-2029 for different targets
EPC, Electric, Gas safety inspections
£900 to £2000 fee

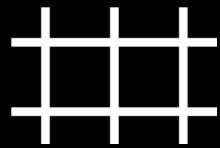
Listings near DLAs





EASTERN SLA (1)

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2016-2021 in St. George West & Eastville
All privately-let properties
EPC, Electric, Gas safety inspections
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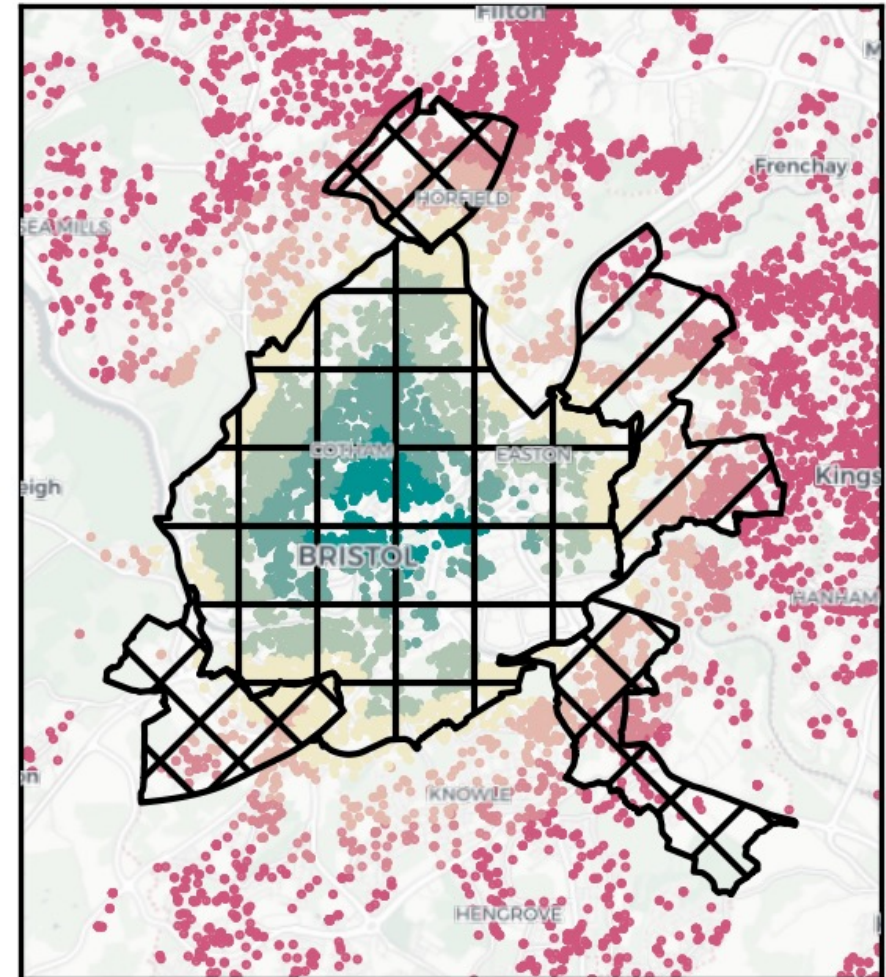
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Listings near DLAs



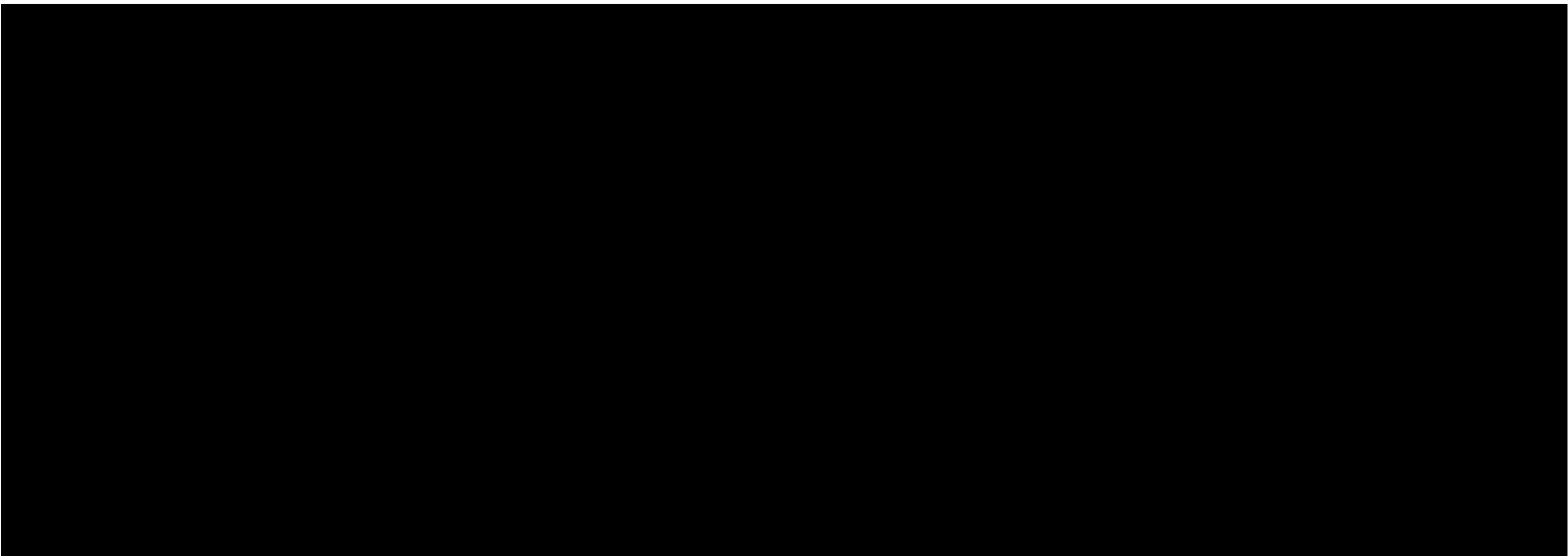


City Council of Bristol (The council)

Housing Act 2004 ("The Act"): Parts 2 and 3

NOTICE OF DESIGNATION OF A CITYWIDE ADDITIONAL LICENSING

NOTICE OF DESIGNATION OF BISHOPSTON AND ASHLEY DOWN, COTHAM AND EASTON WARDS FOR
SELECTIVE LICENSING



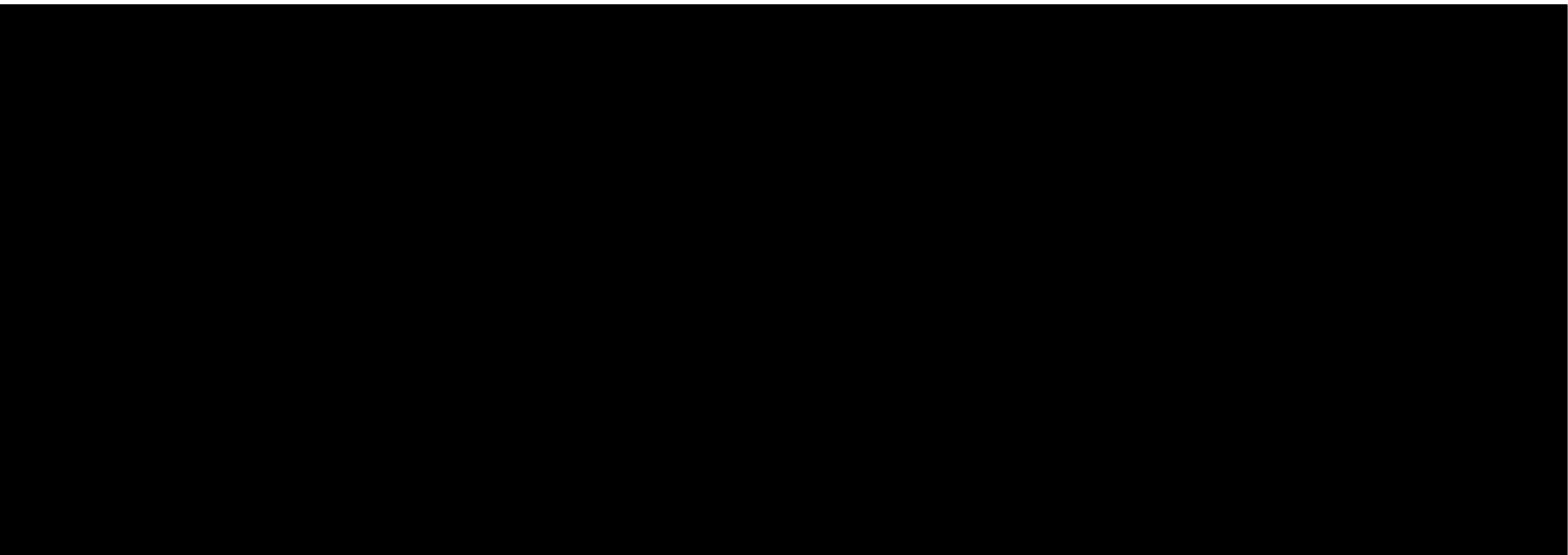


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Bristol landlords oppose new licensing fee scheme



City Council of Bristol (The council)

Housing Act 2004 ("The Act"): Parts 2 and 3

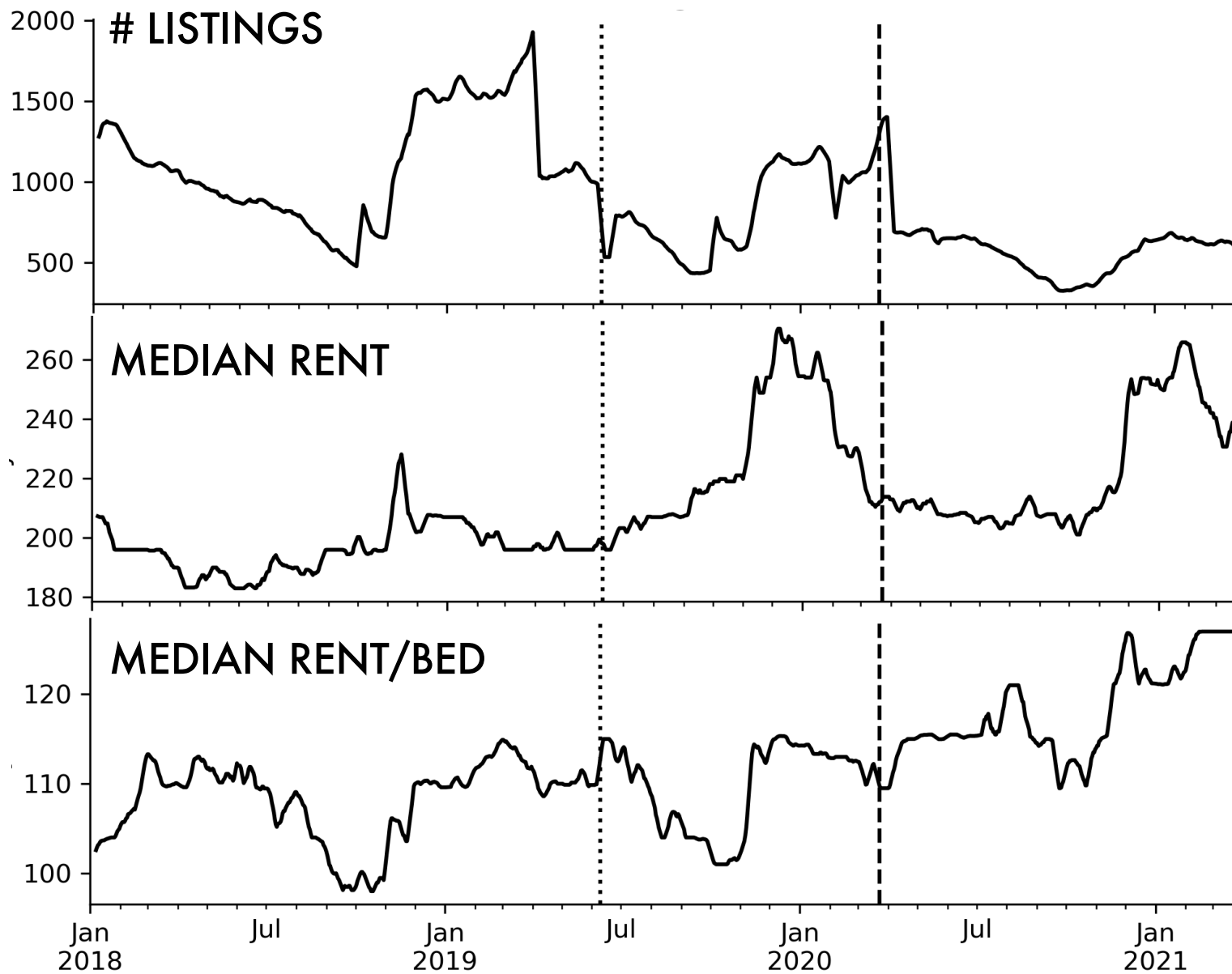
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NOTICE OF DESIGNATION OF **BISHOPSTON AND ASHLEY DOWN, COTHAM AND EASTON WARDS** FOR
SELECTIVE LICENSING



Bristol landlords oppose new licensing fee scheme

Letting agents and landlords criticised the cost of the licences, and also warned the extra fees could lead to higher rents for tenants, less affordable housing getting built, and some landlords leaving the market, meaning fewer homes available to rent.

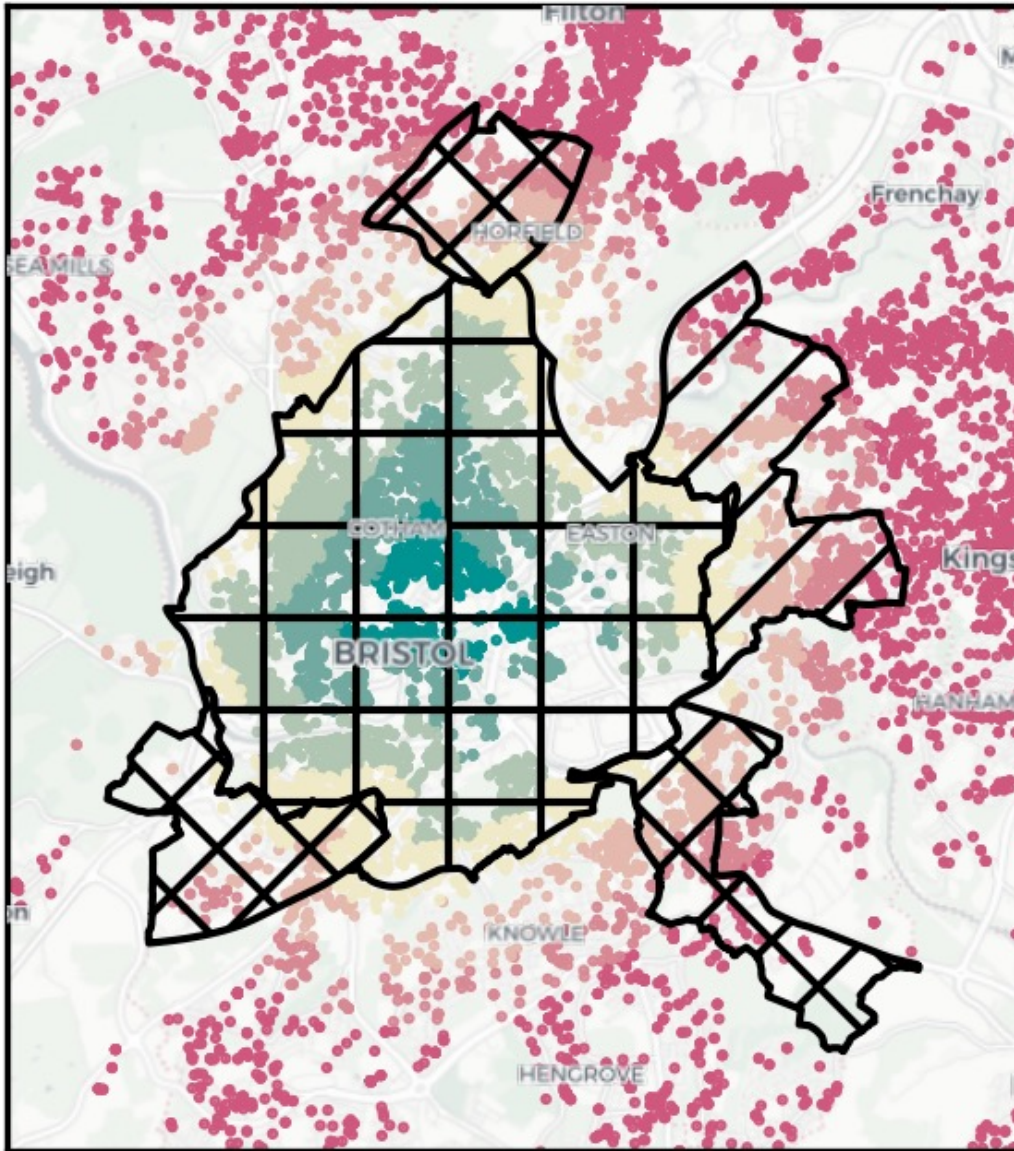


Fewer rentals?

Higher rent?

Less affordable?

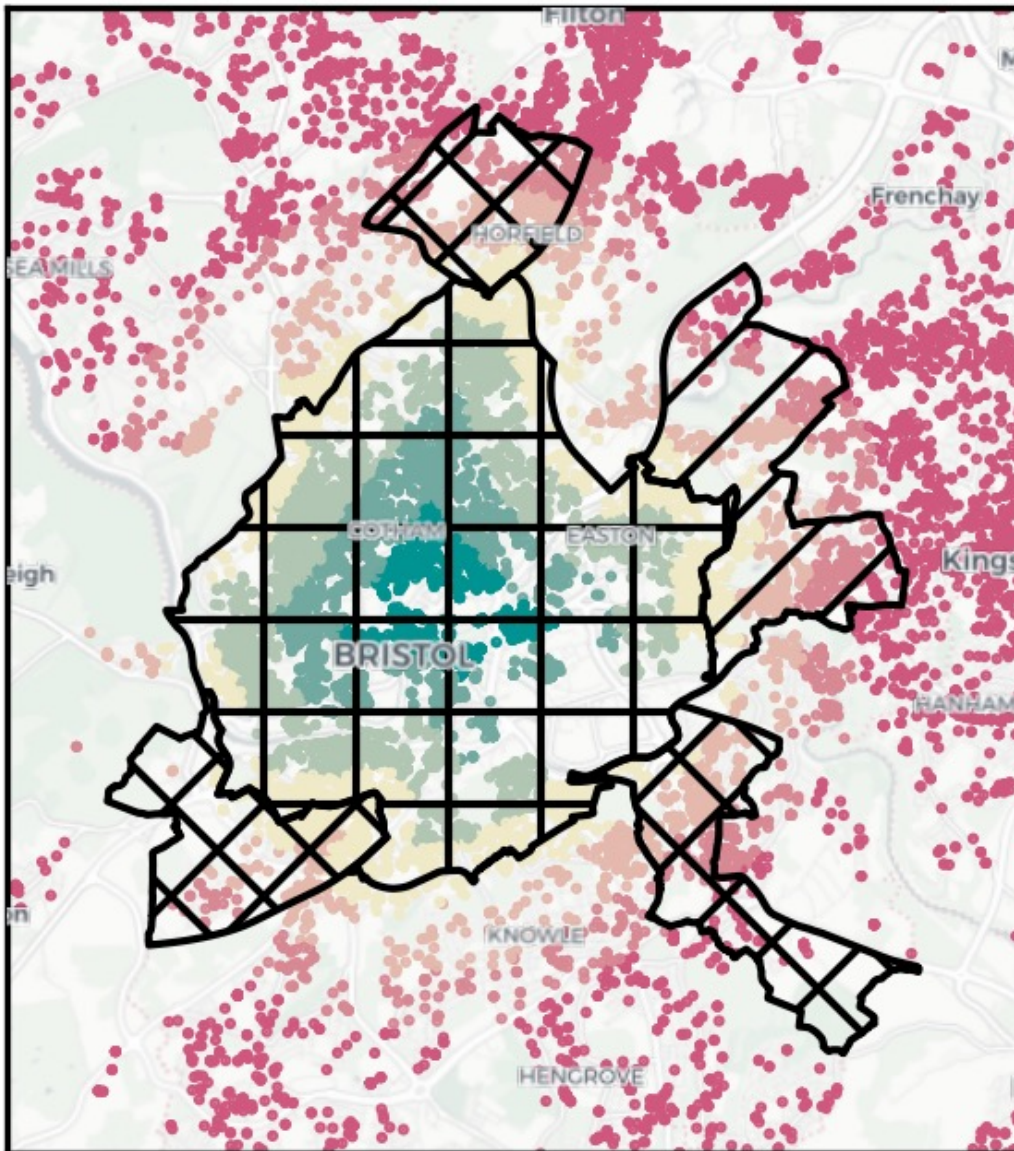
Listings near DLAs



Given the geography of these regulation areas, how can we be sure that their imposition is reflected in the “total” city rates?

Displacements **within** cities matter!

Listings near DLAs



Given the geography of these regulation areas, how can we be sure that their imposition is reflected in the “total” city rates?

Displacements **within** cities matter!

Neighborhood-level rent regulation distorts *local*, not city-wide, incentives.

We want to know: for similar properties, did licensing make a difference *on the margin*?

That is, **at the boundary of the DLA**, can we detect the difference the DLA makes?

WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

PRELIMINARY RESULTS

Treated listings inside the DLA after treatment.

Pre-treated listings inside the DLA before treatment.

Substitute listings outside the DLA but nearby.

Pre-Substitute listings outside of the DLA before treatment.

IDENTIFICATION STRATEGY 1: SDID_c

Treated listings inside the DLA after treatment.

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At the DLA boundary, estimate discontinuity

Pre-treated and **Pre-substitute**, then **Treated** and **Substitute**

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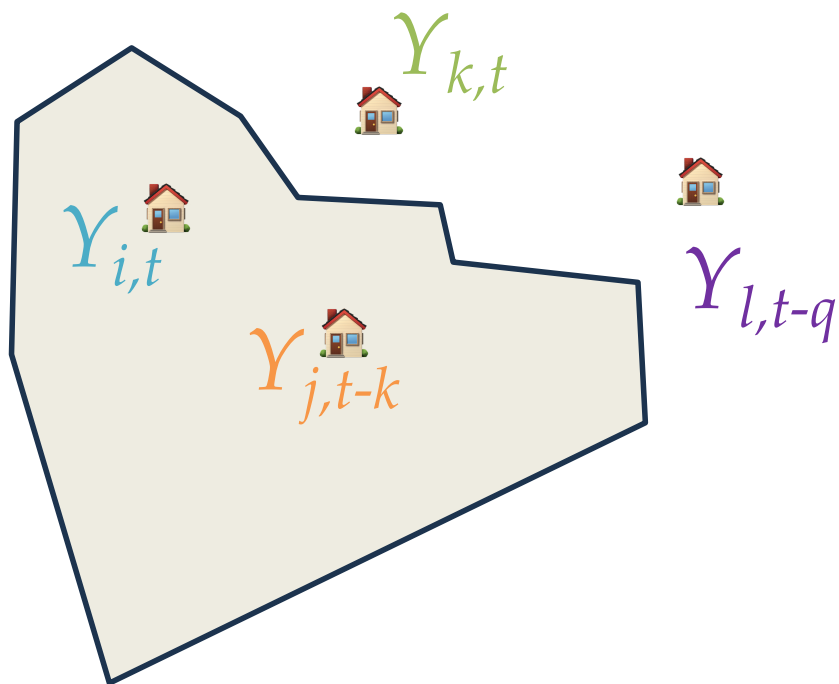
At the DLA boundary, estimate discontinuity

Pre-treated and **Pre-substitute**, then **Treated** and **Substitute**

If the estimated discontinuity at the DLA boundary *changes*,
we have a **spatial difference in discontinuities**.

ATE: difference in estimated boundary discontinuities

IDENTIFICATION STRATEGY 1: SDID_c



**Borrow counterfactual from
*just over the boundary***

Pretreatment model:

fit on Y_j POs from Y_l

Treatment model:

fit on Y_i POs from Y_k

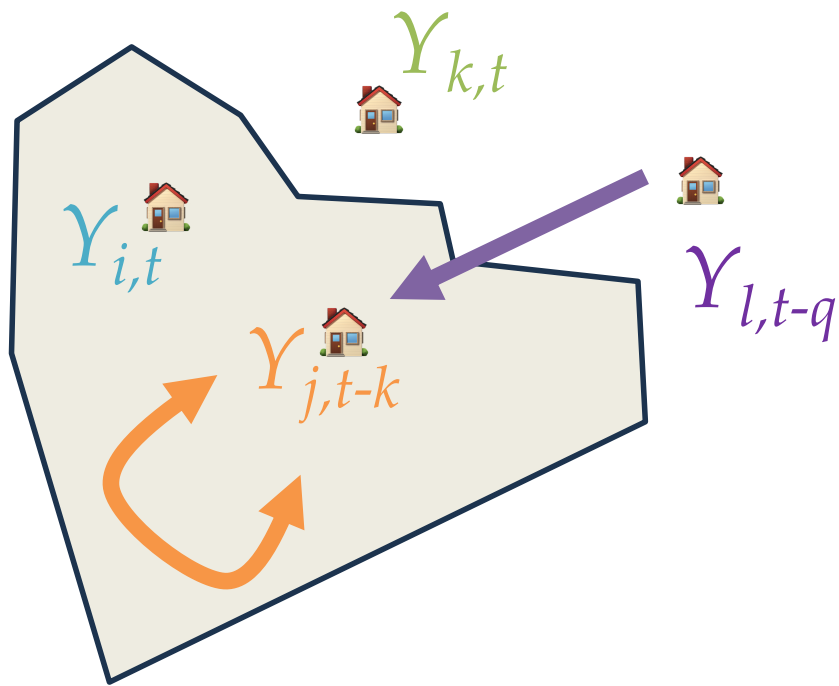
Substitute model:

fit on Y_k POs from Y_i

Pre-substitute model:

fit on Y_l POs from Y_j

IDENTIFICATION STRATEGY 1: SDID_c



Pretreatment model:
fit on Y_j POs from Y_l

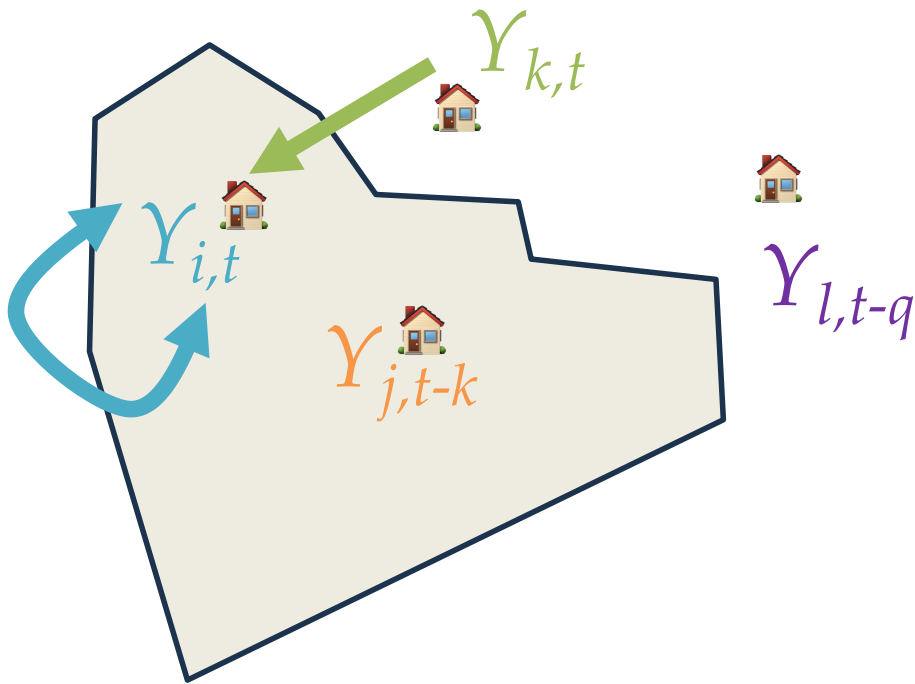
Treatment model:
fit on Y_i POs from Y_k

Substitute model:
fit on Y_k POs from Y_i

Pre-substitute model:
fit on Y_l POs from Y_j

Discontinuity: predicted outcome under “pre-treated” inside model vs. “never-treated” outside model

IDENTIFICATION STRATEGY 1: SDID_c



Pretreatment model:
fit on Y_j POs from Y_l

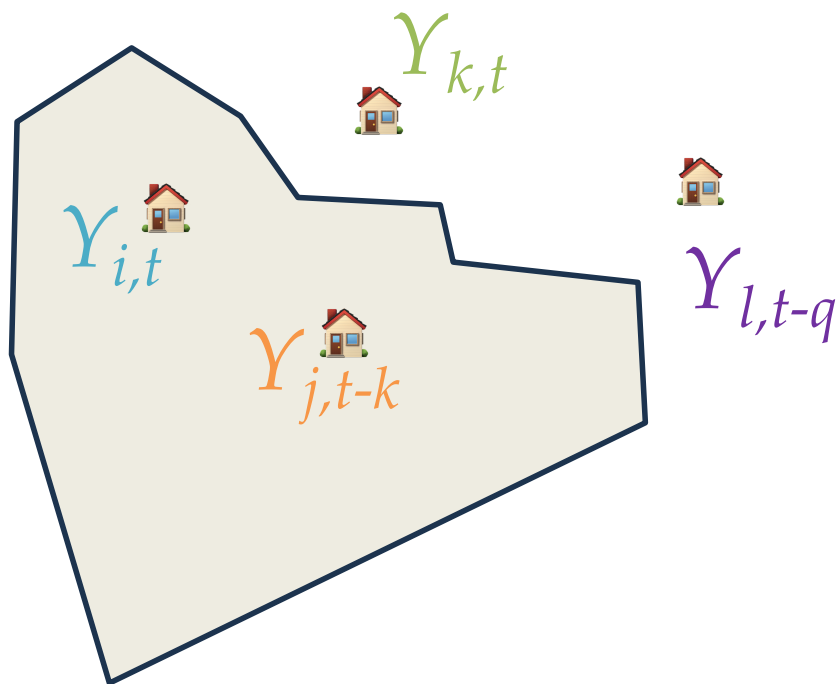
Treatment model:
fit on Y_i POs from Y_k

Substitute model:
fit on Y_k POs from Y_i

Pre-substitute model:
fit on Y_l POs from Y_j

Discontinuity: predicted outcome under “treated” inside model vs. “never-treated” outside model

IDENTIFICATION STRATEGY 1: SDID_c



ATT: average discount. for top 2

ATE: average discount. overall

Pretreatment model:

fit on Y_j POs from Y_l

Treatment model:

fit on Y_i POs from Y_k

Substitute model:

fit on Y_k POs from Y_i

Pre-substitute model:

fit on Y_l POs from Y_j

IDENTIFICATION STRATEGY 1: SDID_c

Treated listings inside the DLA after treatment.

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Substitute listings outside the DLA but nearby.

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IDENTIFICATION STRATEGY 2: QUARTET.+ X

Treated listings inside the DLA after treatment.

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Substitute listings outside the DLA but nearby.

Pre-Substitute listings outside of the DLA before treatment.

Match:

1. treated to predecessor—match back in time w/in DLA
2. treated to substitute—match contemporary to out of DLA
3. substitute to predecessor—match back in time out of DLA

IDENTIFICATION STRATEGY 2: QUARTET + X

Treated listings inside the DLA after treatment.

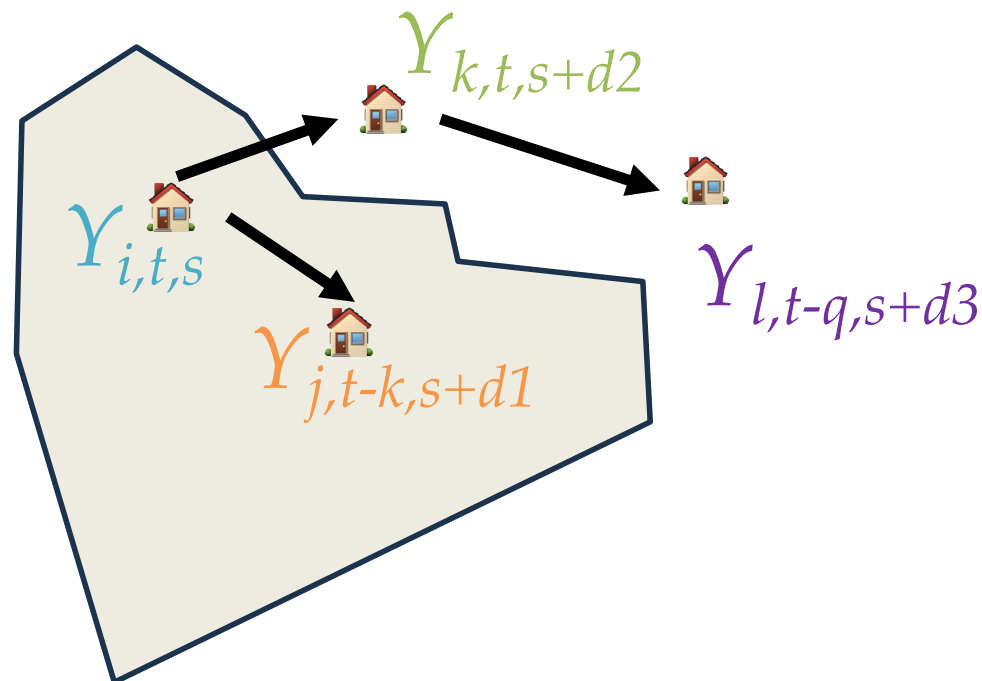
Pre-treated listings inside the DLA before treatment.

Substitute listings outside the DLA but nearby.

Pre-Substitute listings outside of the DLA before treatment.

Match:

1. **treated** to **predecessor**
2. **treated** to **substitute**
3. **substitute** to **predecessor**



IDENTIFICATION STRATEGY 2: QUARTET + X

Treated listings inside the DLA after treatment.

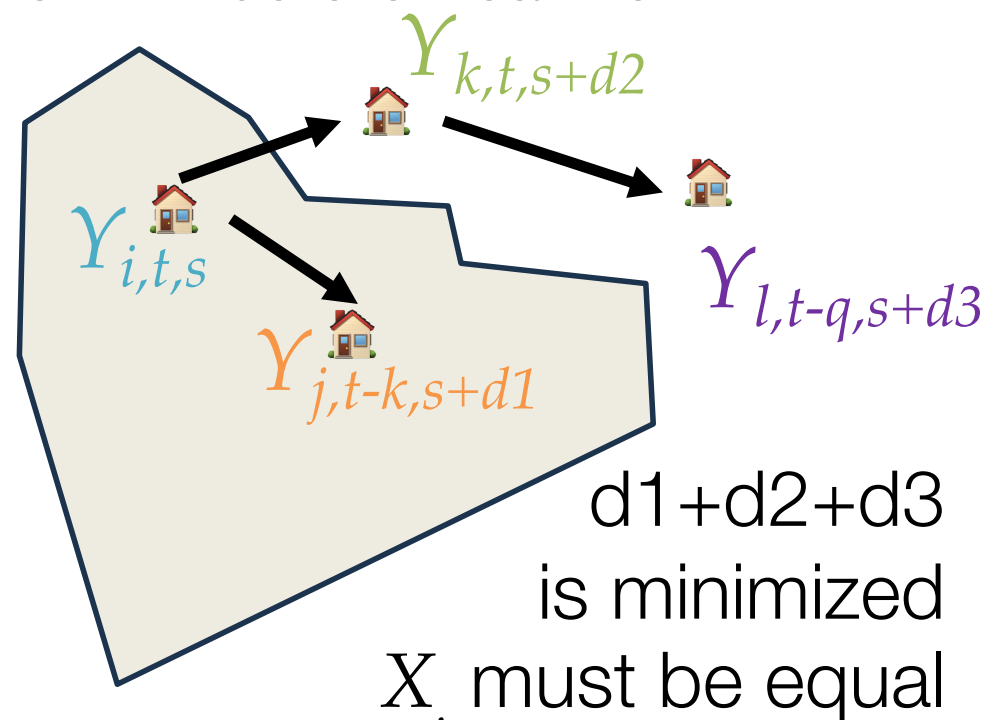
Pre-treated listings inside the DLA before treatment.

Substitute listings outside the DLA but nearby.

Pre-Substitute listings outside of the DLA before treatment.

Match:

1. **treated** to **predecessor**
2. **treated** to **substitute**
3. **substitute** to **predecessor**



IDENTIFICATION STRATEGY 2: QUARTET + X

minimize $\sum d_{ij}t_{ij} + \sum d_{ik}s_{ik} + \sum d_{kl}p_{kl}$
subject to

$$\sum_i t_{ij} = 1$$

all treatments get one pretreatment

$$\sum_i s_{ik} = 1$$

all treatments get one substitute

$$\sum_l p_{kl} \geq \sum_j s_{jk}$$

every selected substitute gets a presub

$$\sum_k p_{kl} \leq 1$$

presubs can be used at most once

IDENTIFICATION STRATEGY 2: QUARTET + X

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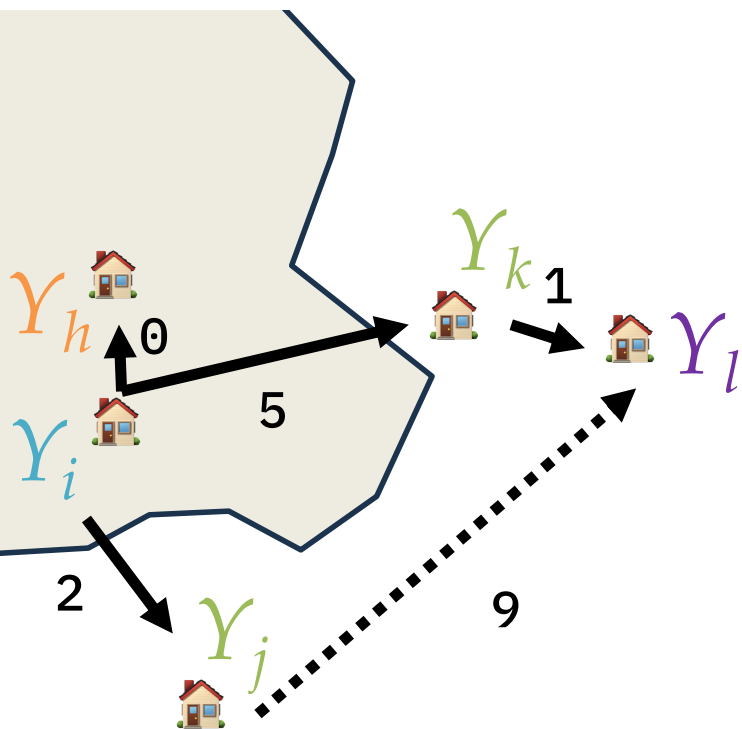
every selected substitute gets a presub

$$\sum_k p_{kl} \leq 1$$

presubs can be used at most once

Separable subproblems! *Select nearest predecessor to treatment,*
then *select sub+presub that minimize sum of distance*

IDENTIFICATION STRATEGY 2: QUARTET + X

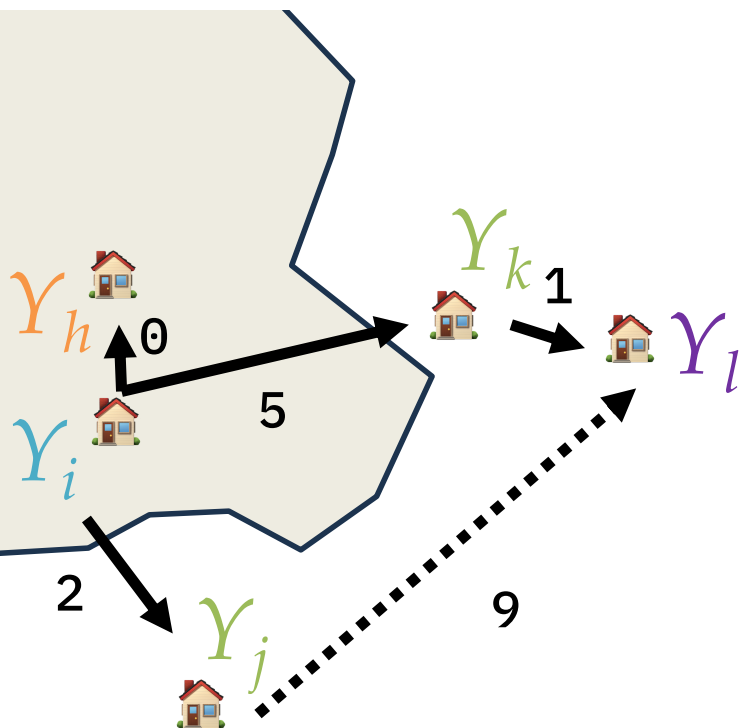


Step by step optimization
greedily picks $\langle hijl \rangle$
for a cost of 11

Joint optimization over subs
and presubs picks $\langle hikl \rangle$
for a cost of 6!

Indeed, j might go unused
w/o a nearer presub!

IDENTIFICATION STRATEGY 2: QUARTET + X



Step by step optimization
greedily picks $\langle hijl \rangle$
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Joint optimization over subs
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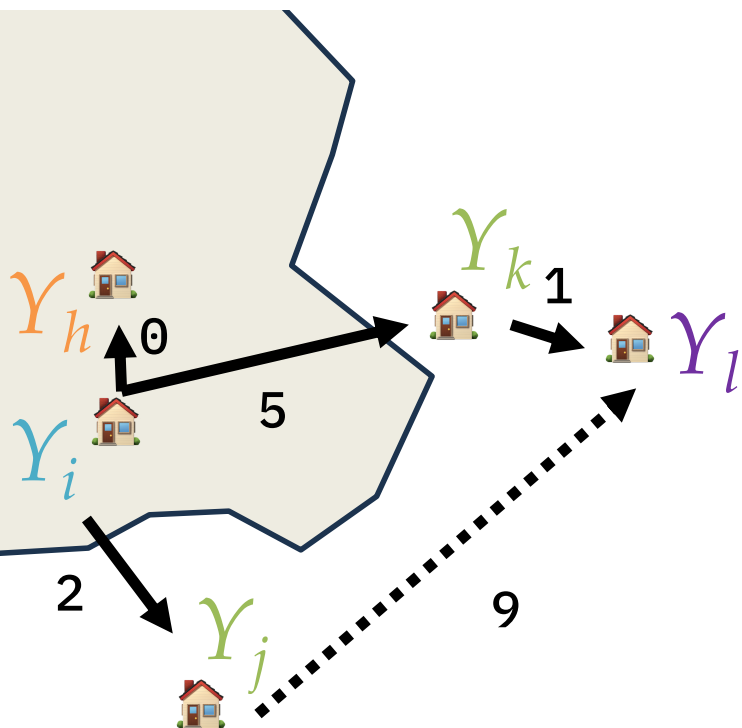
**Blows up the
assignment problem...**

$$2(n_t n_u) \rightarrow n_t n_u (n_u - 1)$$

So we'll stay w/ greedy

Indeed, j might go unused
w/o a nearer presub!

IDENTIFICATION STRATEGY 2: QUARTET + X



Step by step optimization
greedily picks $\langle h i j l \rangle$
for a cost of 11

Joint optimization over subs
and presubs picks $\langle h i k l \rangle$
for a cost of 6!

**Blows up the
assignment problem...**

176,000,000 $\rightarrow$ 968,000,000,000

So we'll stay w/ greedy

Indeed, j might go unused
w/o a nearer presub!

IDENTIFICATION STRATEGY 2: QUARTET + X

$$Y_i = Y_h + X_i + e_i$$

TREATED AREA MODEL

$$Y_k = Y_l + X_k + e_k$$

UNTREATED AREA MODEL

$$\tau_1 = \hat{Y}_0 - Y_1 = Y_h + X + u$$

TREATED PO MODEL

$$\tau_0 = Y_0 - \hat{Y}_1 = Y_l + X + u$$

UNTREATED PO MODEL

$$t \sim Y + X$$

PROPENSITY MODEL

$$\text{ATE: } n^{-1} \sum \tau_1 p + \tau_0 (1-p)$$

IDENTIFICATION STRATEGY 2: QUARTET + X

All models are GWRFs using rental characteristics (about 125 nearest neighbors; X-learner propensity model uses 1000-meter fixed bandwidth)

All inference uses residual bootstrap

Additionally, for X-learner we also shuffle the propensity score separately

ESTIMATION & INFERENCE

WHAT & WHY ARE DLA?

IDENTIFICATION & ESTIMATION

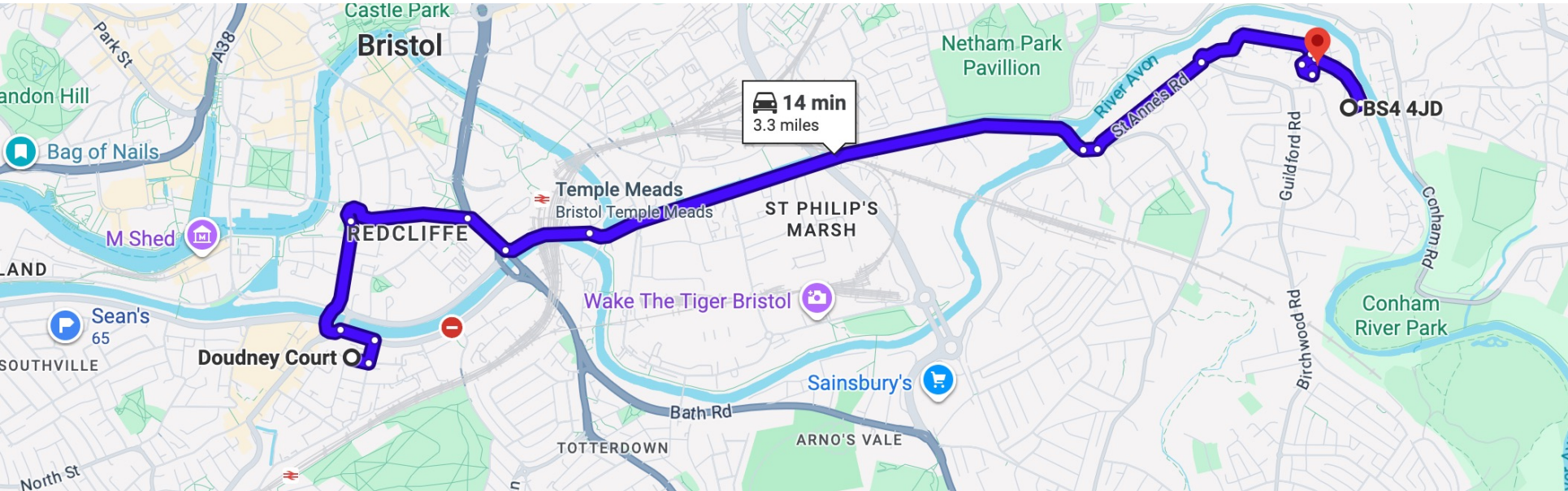
PRELIMINARY RESULTS

SLA OR NAY?

kind	uid	Treated	In DLA?	Date	Postcode	Beds	Baths	Rent
treated	9cae7e1edb	TRUE	TRUE	2019-08-14	BS34AP	2	0	242
pretreat	6f91be8c67	FALSE	TRUE	2018-08-16	BS34AP	2	0	230
substitute	573f5484f2	FALSE	FALSE	2019-06-03	BS44JD	2	0	207
presubstitute	70cfae8c52	FALSE	FALSE	2018-01-07	BS44AU	2	0	167

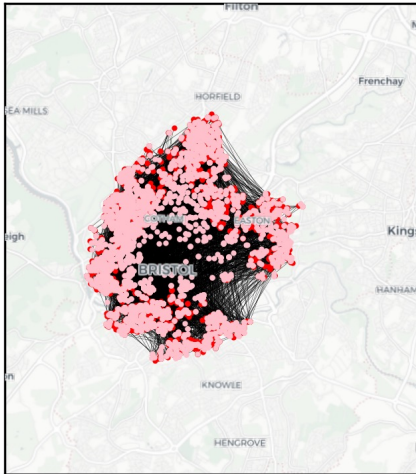
MATCH EXAMPLE

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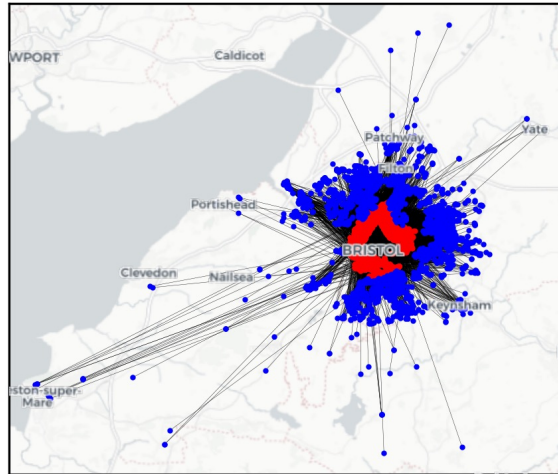


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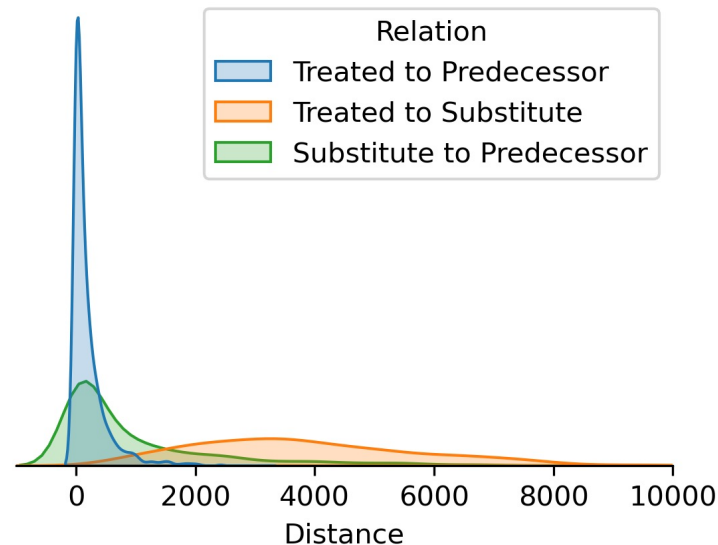
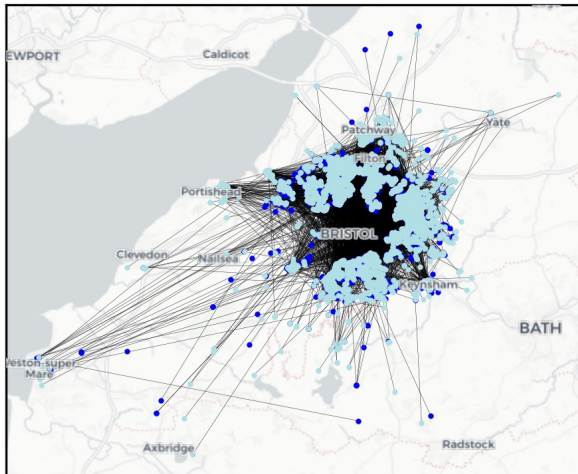
Treated to Predecessor



Treated to Substitute



Substitute to Predecessor



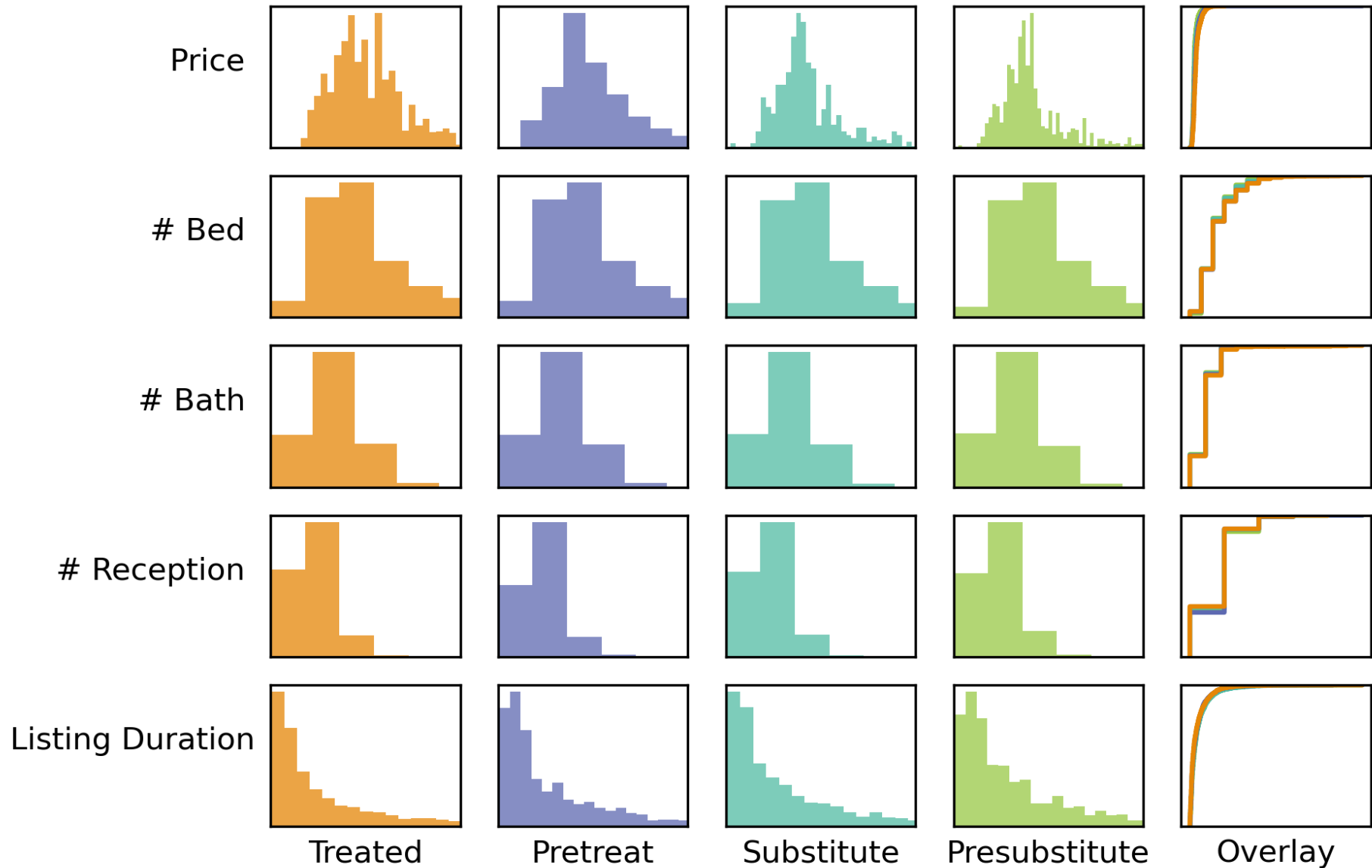
Proximity of the step-wise matching leaves something to be desired... but!

listings are 2018-2022
policy is 2019-2024

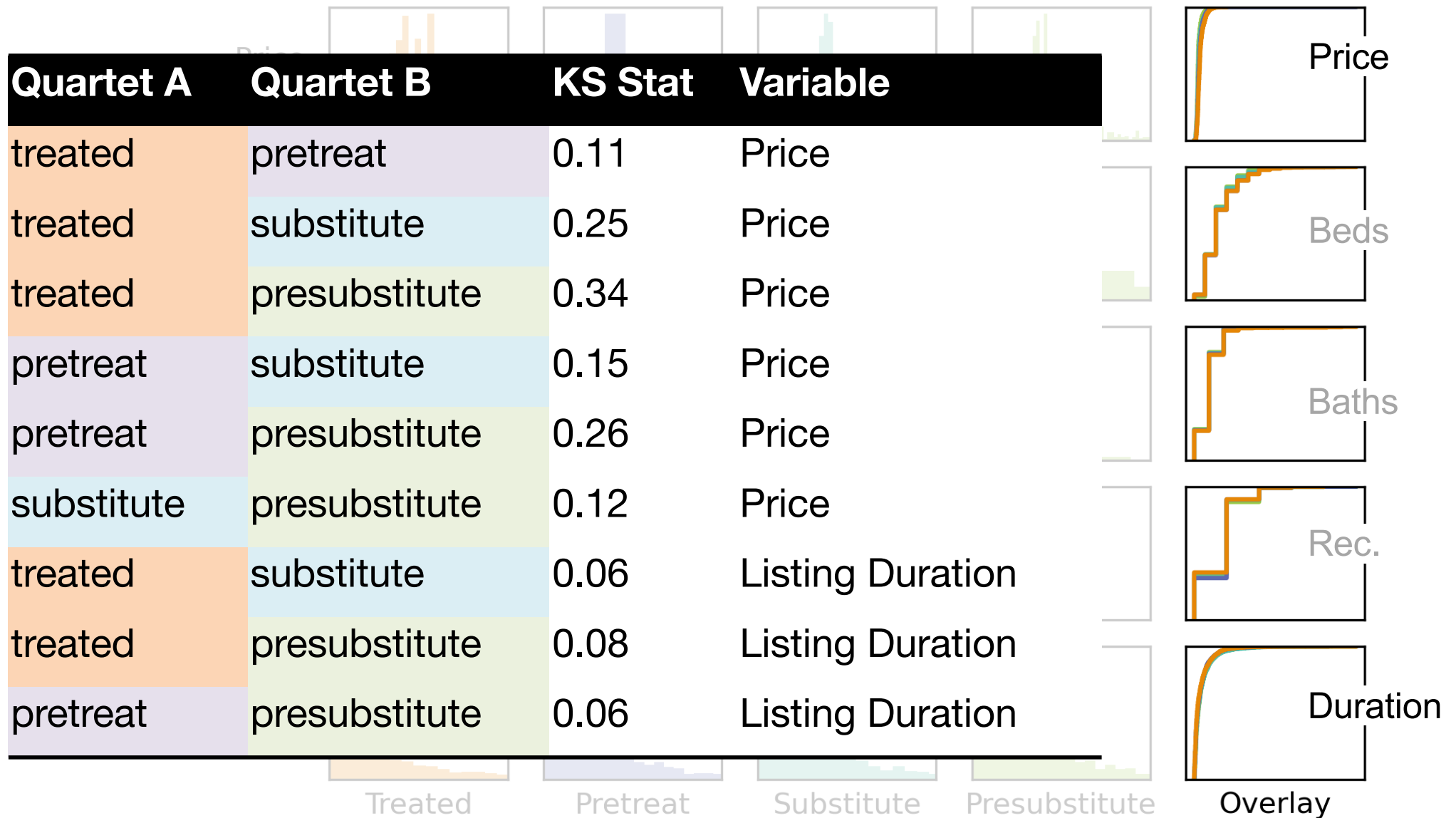
2 more years of data for
substitutes won't
necessarily improve this a
large amount.

Computation is still running
for the joint match.

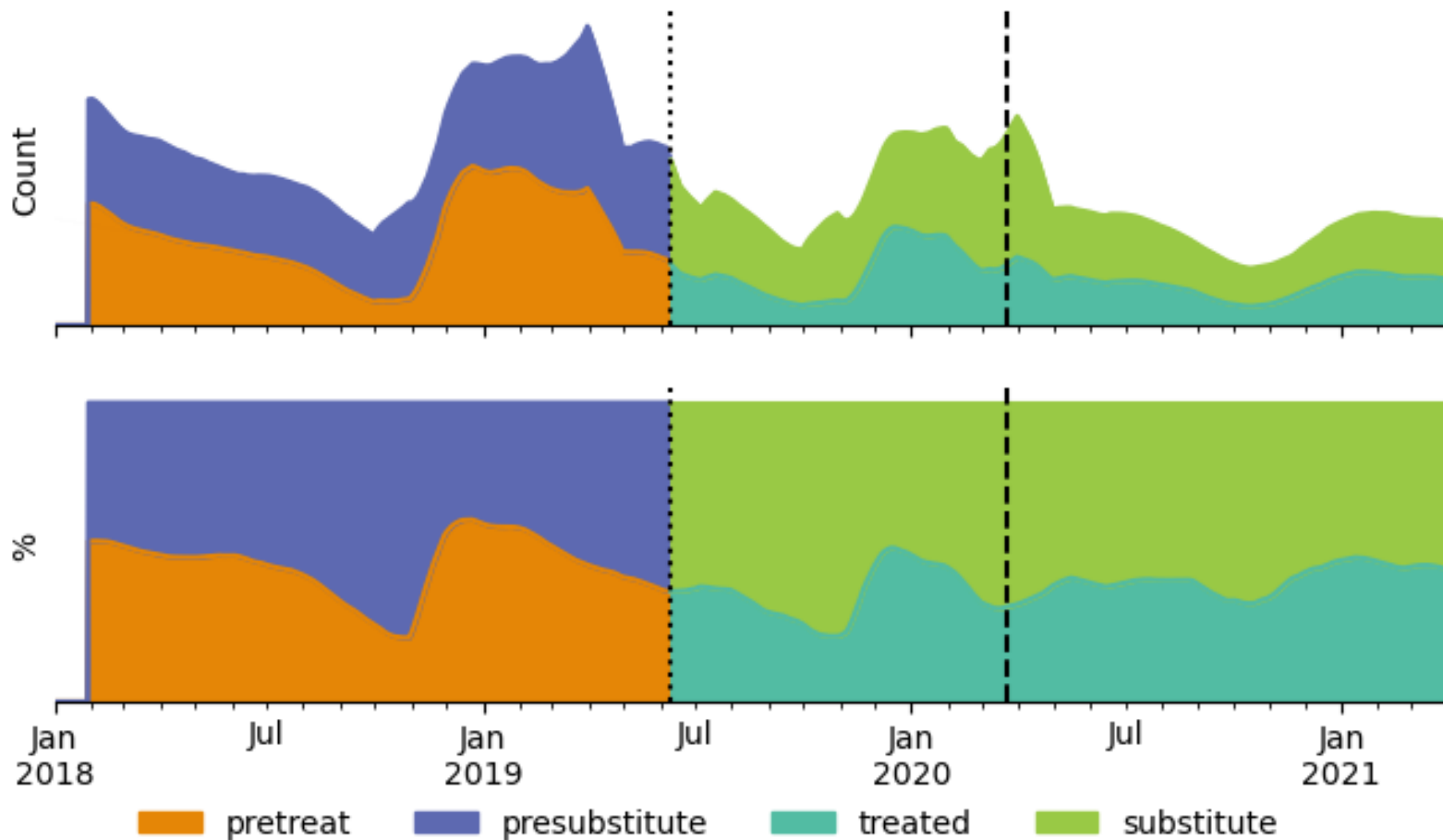
MATCH QUALITY



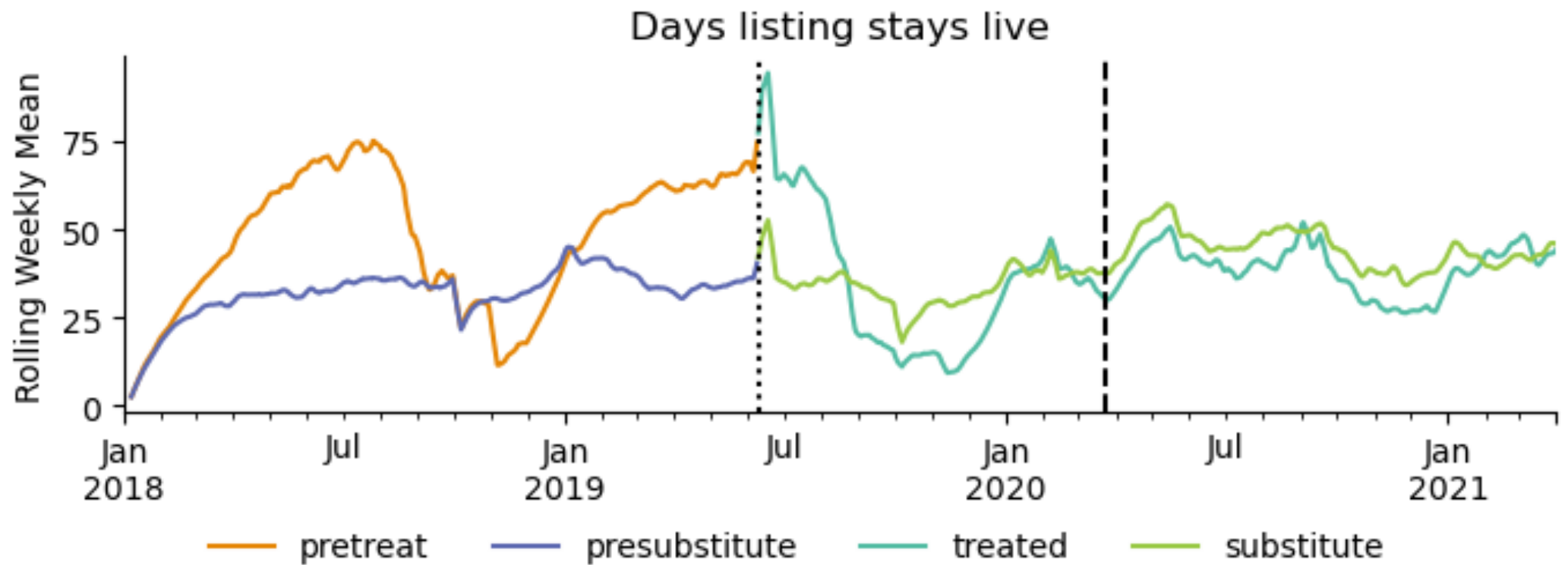
MATCHES ARE VERY BALANCED



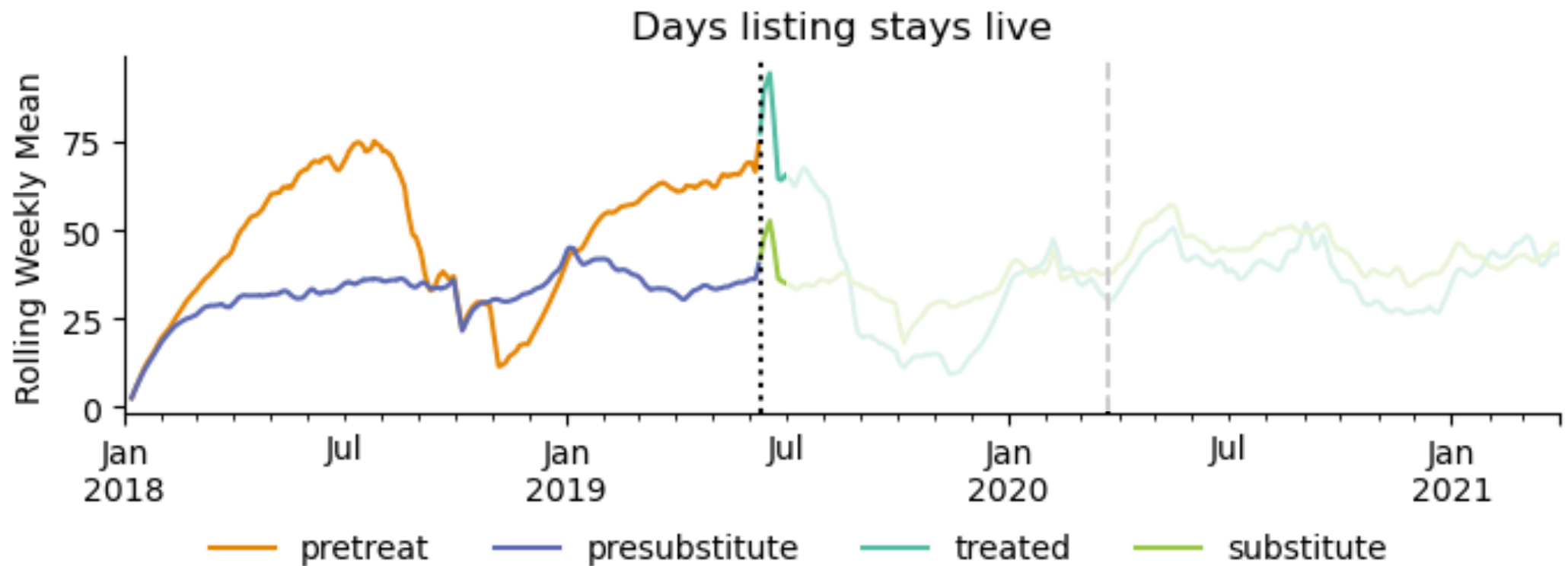
MATCHES ARE VERY BALANCED ON EXOG



SUPPLY EFFECTS: TOTAL LISTINGS

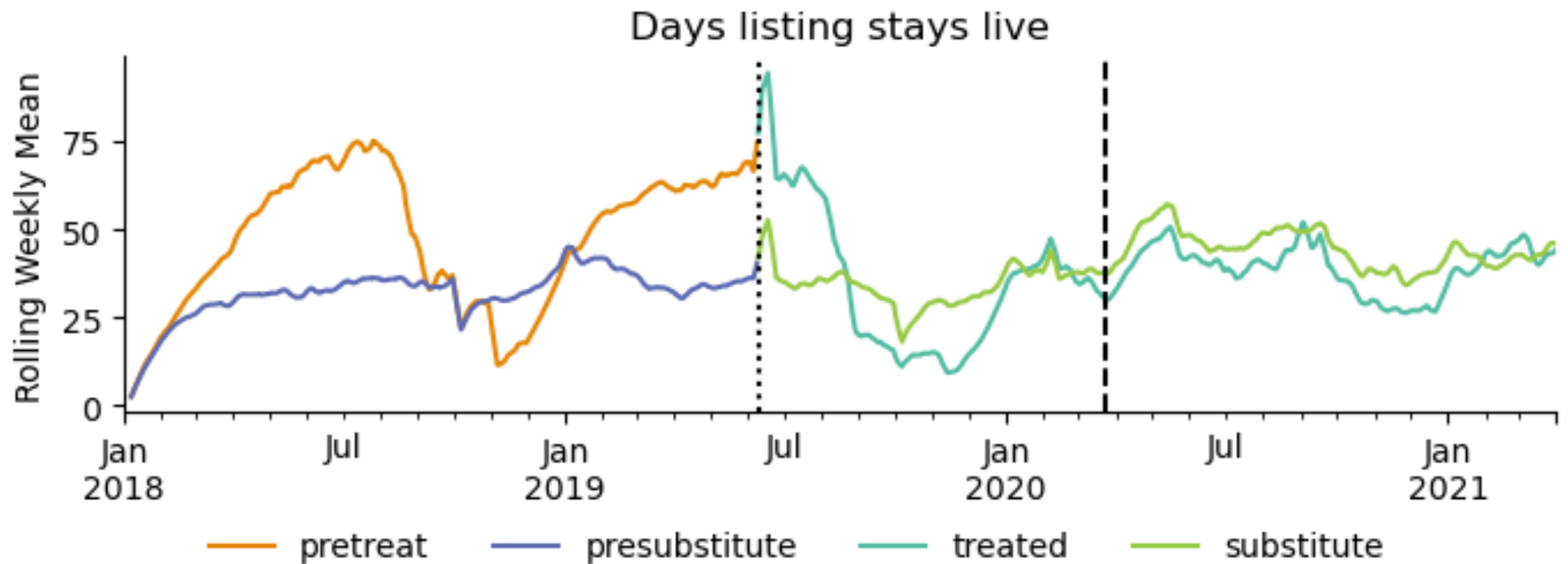


SUPPLY EFFECTS: STALENESS



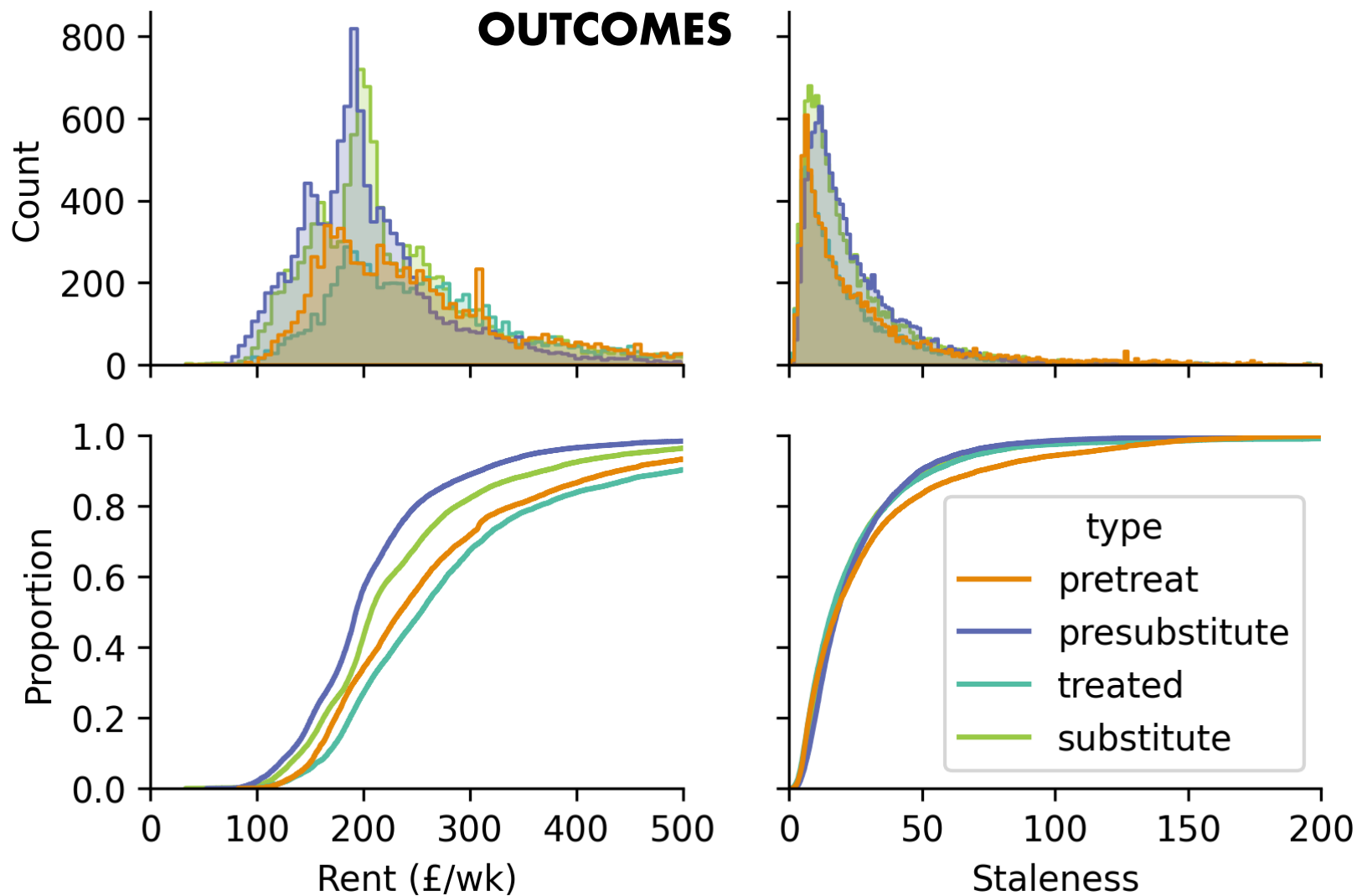
Immediately after policy imposition, there's a bump that's not apparent in July 2018 or 2020, but...

SUPPLY EFFECTS: STALENESS

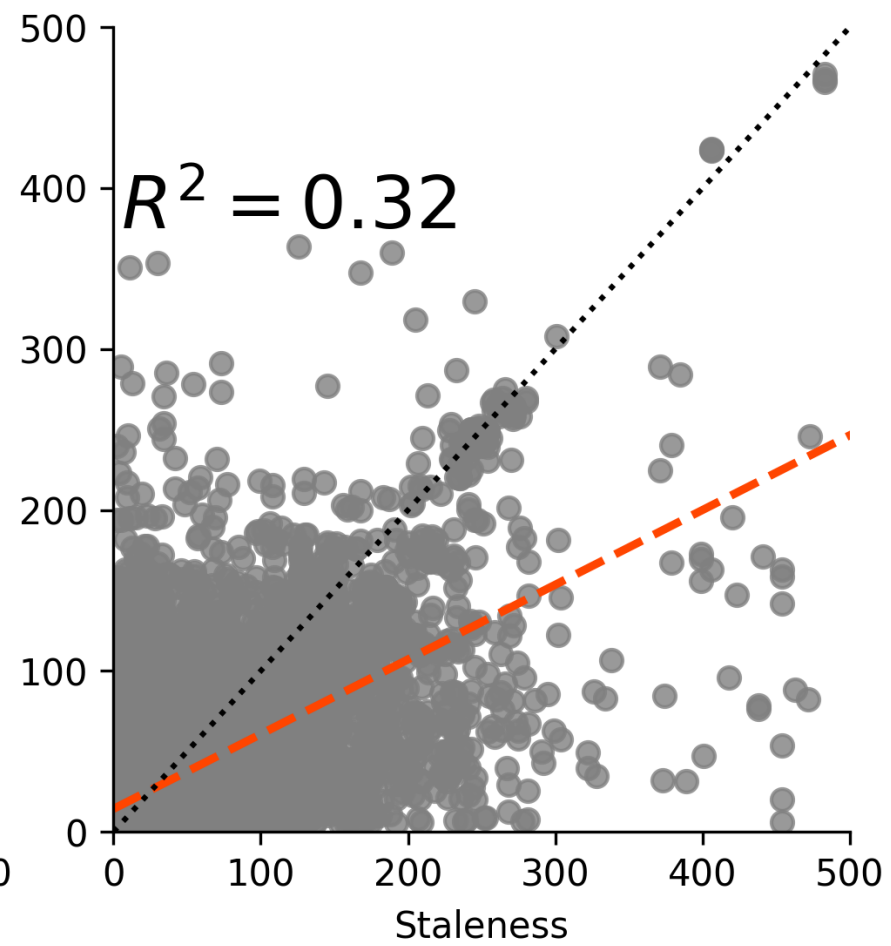
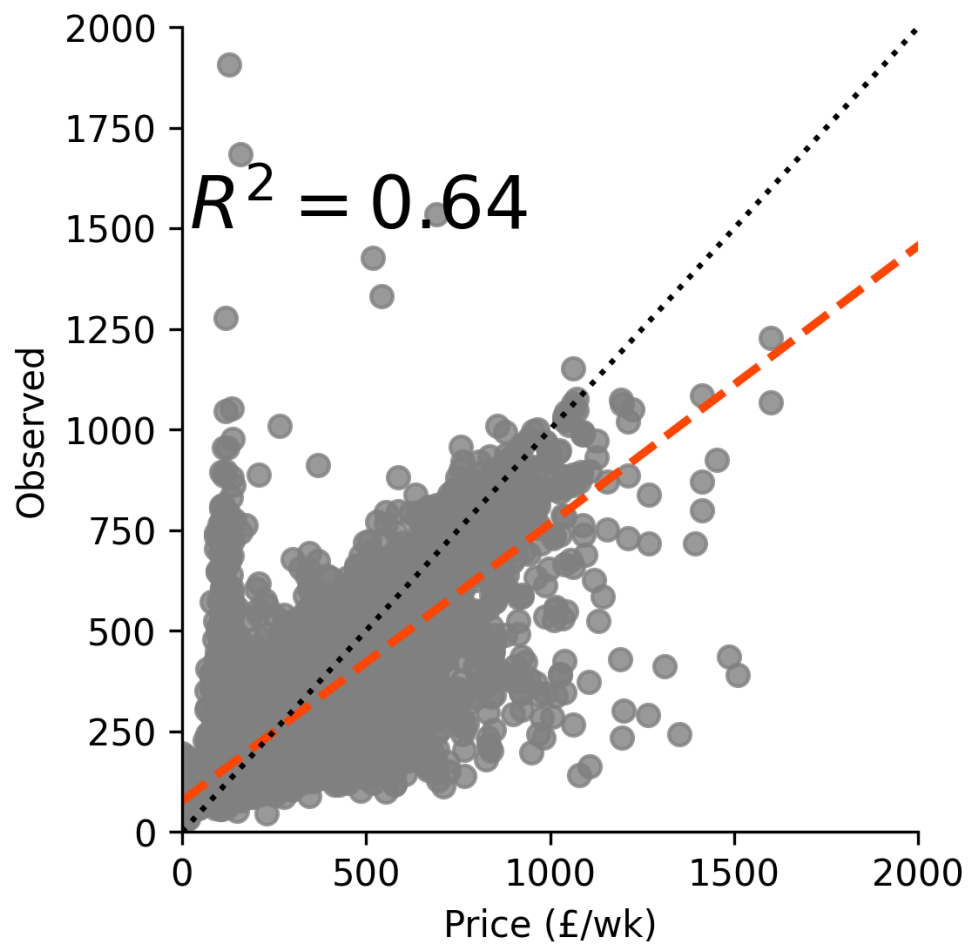


Immediately after policy imposition, there's a bump that's not apparent in July 2018 or 2020, but... certainly does not persist!

SUPPLY EFFECTS: STALENESS



SPATIAL DIFFERENCE IN DISCONTINUITY



SPATIAL DIFFERENCE IN DISCONTINUITY

	Overall		Boundary	
outcome	duration	price	duration	price
presubstitute	-4.60	37.03	2.27	17.30
pretreat	-4.04	-26.15	0.38	-20.49
substitute	-2.51	-42.05	6.33	-7.98
treated	4.48	22.07	-0.83	14.63

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Difference between predicted pre-treatment price made using local models outside the DLA (actual) vs models inside the DLA (counterfactual)

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Models trained just inside the DLA boundary predict higher prices for the same houses...

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Models trained just inside the DLA boundary predict higher prices for the same houses... which flips post-treatment.

SPATIAL DIFFERENCE IN DISCONTINUITY

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ATE	1.41	127.31	7.39	60.42

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-25

35

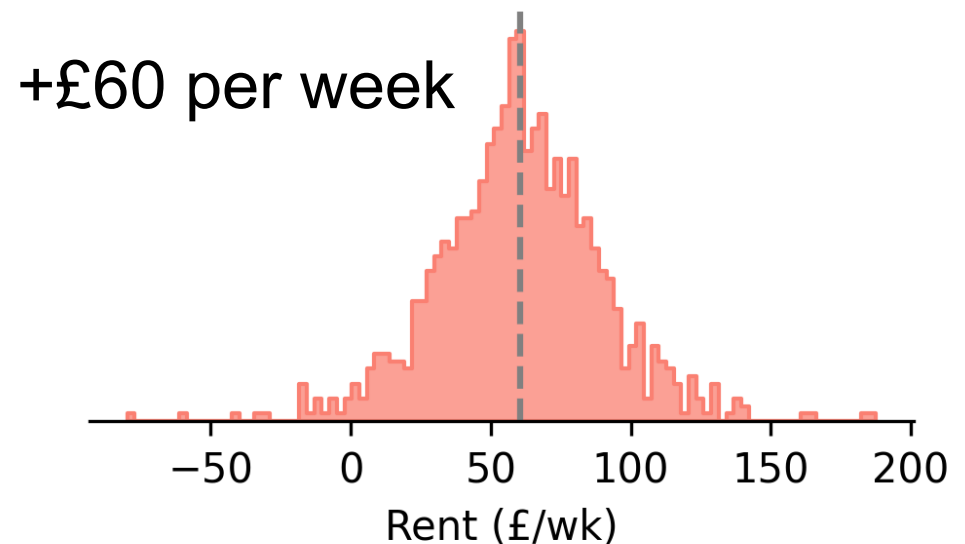
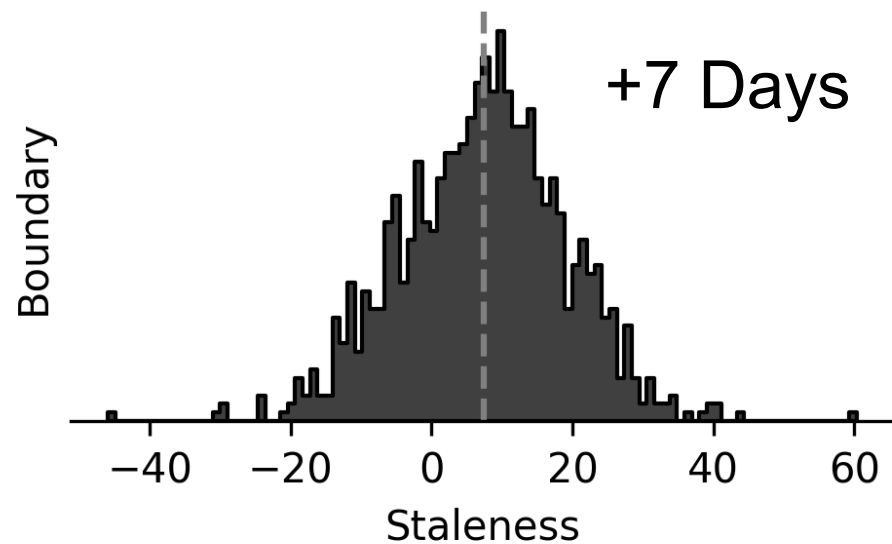
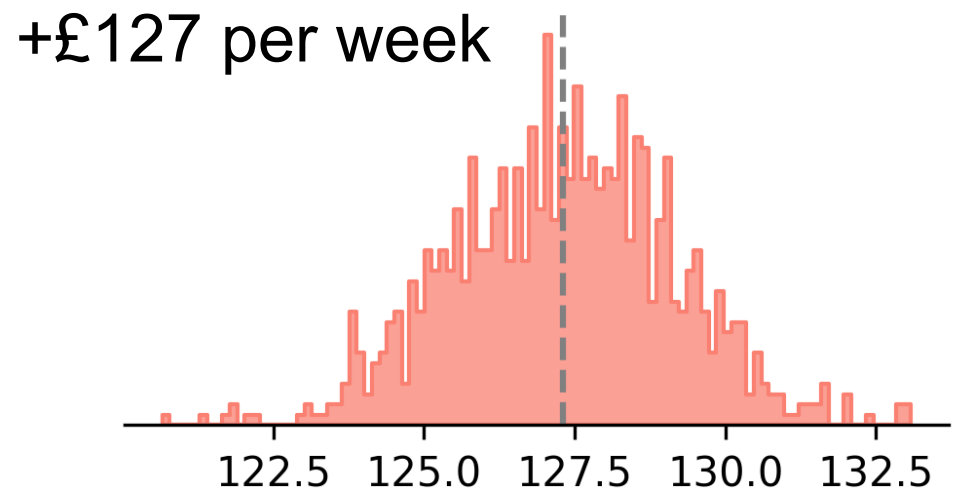
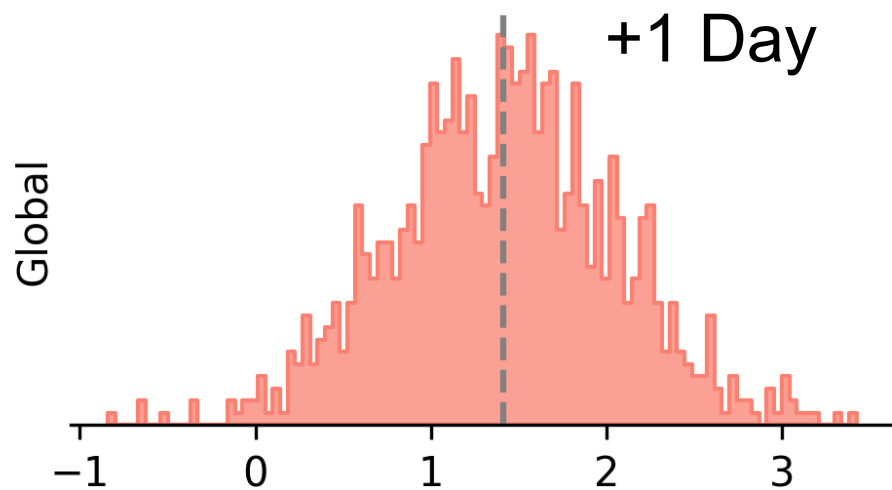
SIZE OF THE GAP FLIP				
outcome	Overall duration	price	Boundary duration	price
ATE	1.41	127.31	7.39	60.42

SPATIAL DIFFERENCE IN DISCONTINUITY

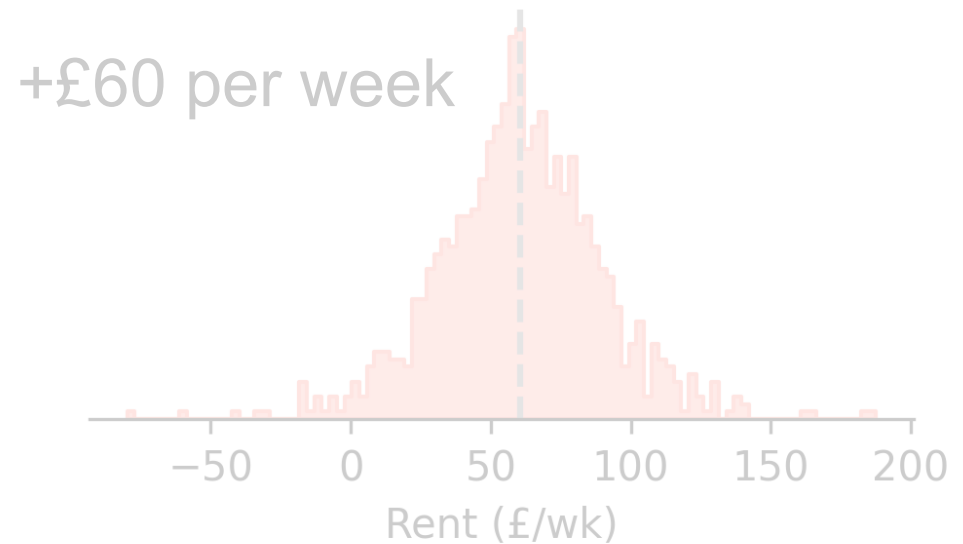
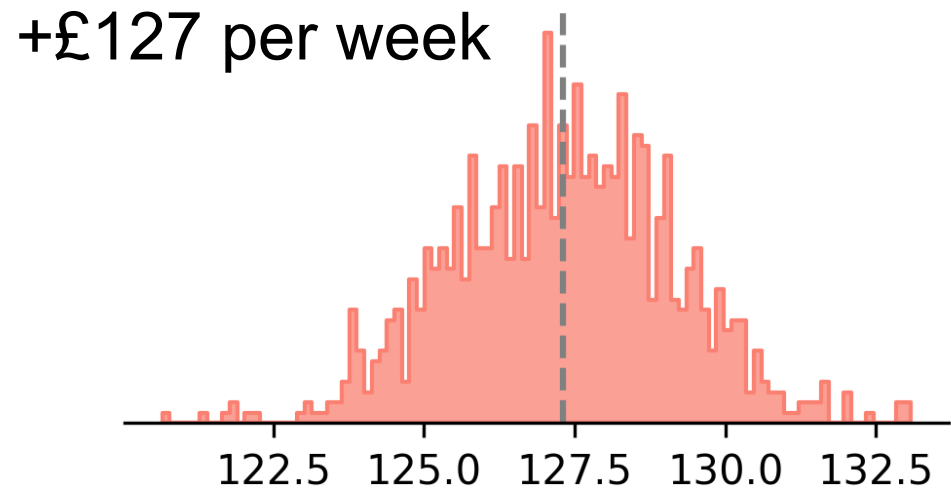
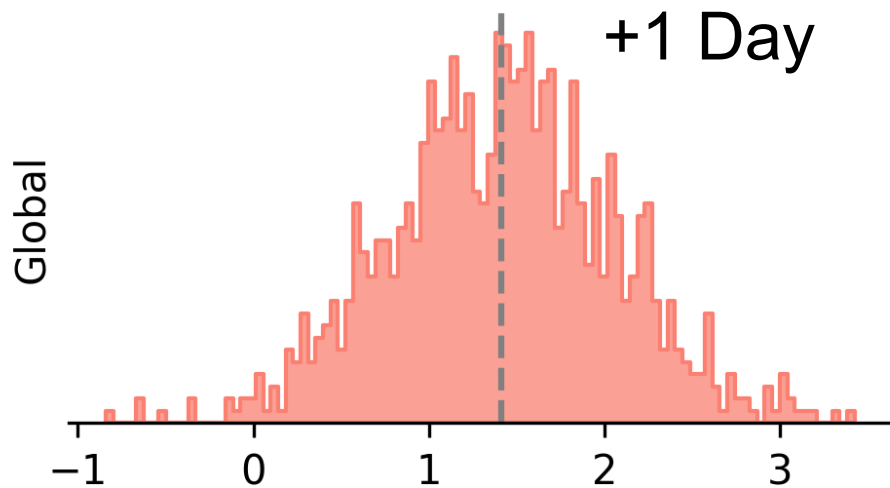
	Overall		Boundary	
outcome	duration	price	duration	price
presubstitute	-4.60	37.03	2.27	17.30
pretreat	-4.04	-26.15	0.38	-20.49
substitute	-2.51	-42.05	6.33	-7.98
treated	4.48	22.07	-0.83	14.63

	Overall		Boundary	
outcome	duration	price	duration	price
ATE	1.41	127.31	7.39	60.42

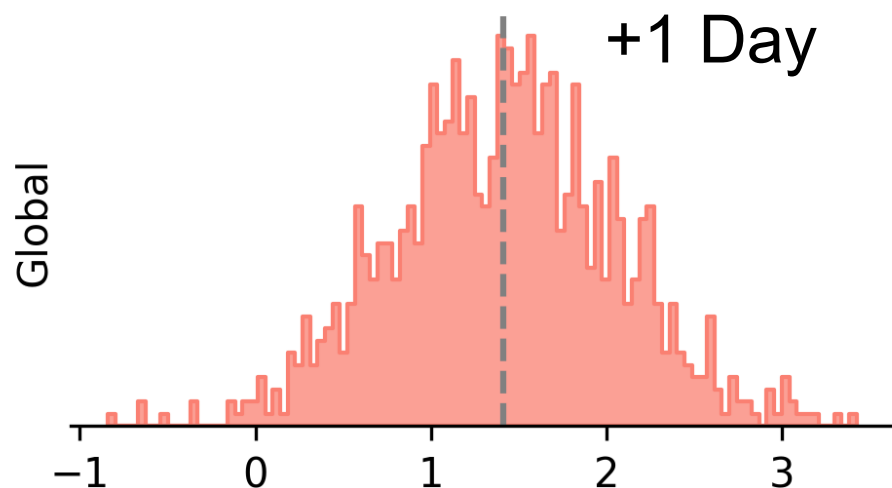
SDID: HIGHER PRICES, SLOWER CLEARING



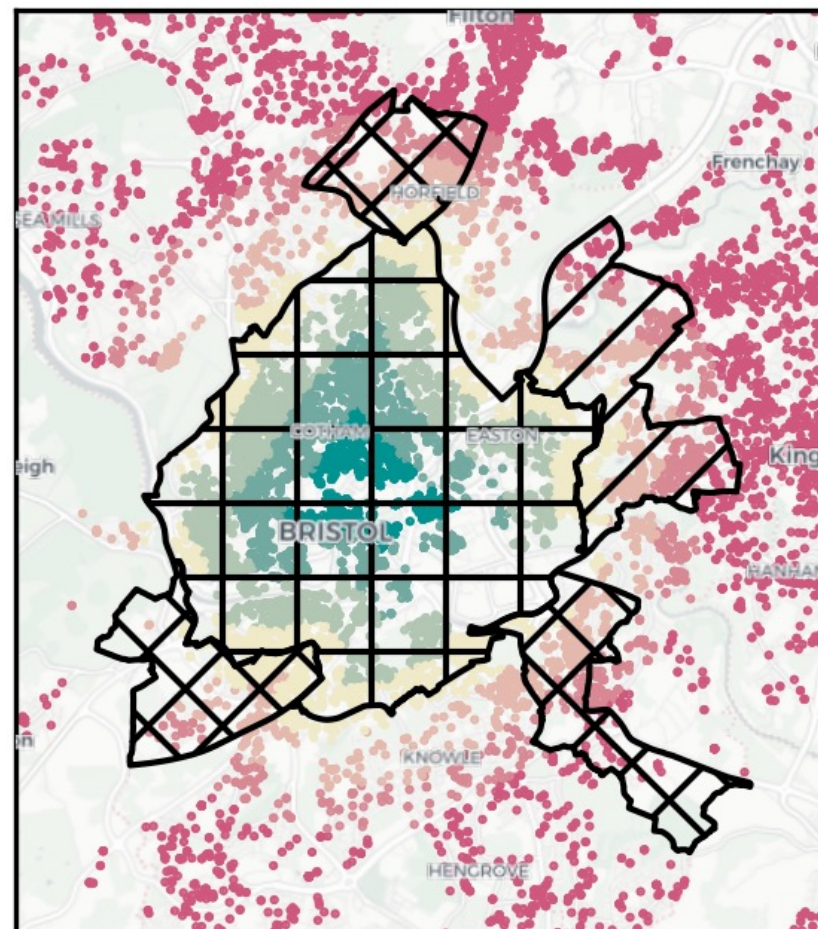
SPATIAL DIFFERENCE IN DISCONTINUITY



SPATIAL DIFFERENCE IN DISCONTINUITY



Listings near DLAs



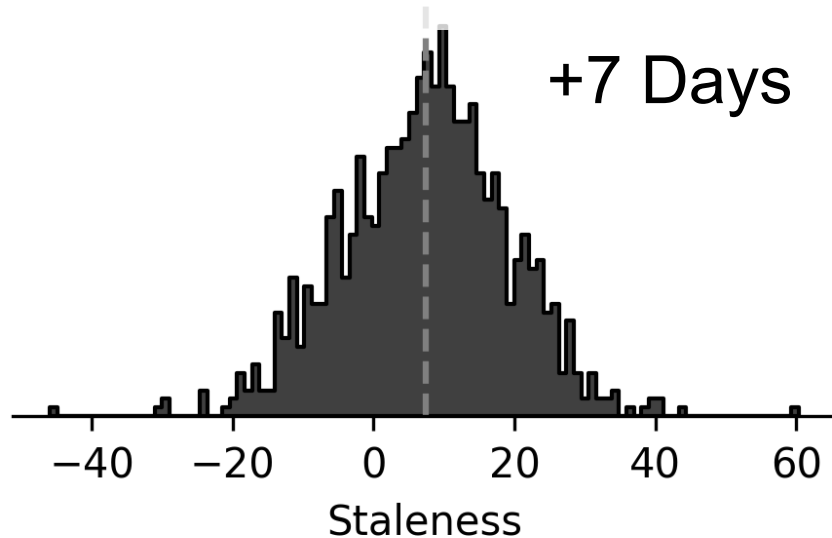
SPATIAL DIFFERENCE IN DISCONTINUITY

FOCUSING NEAR THE BOUNDARY, NO CLEAR CLEARING CHANGE, **RENT CHARGED STILL RISES**

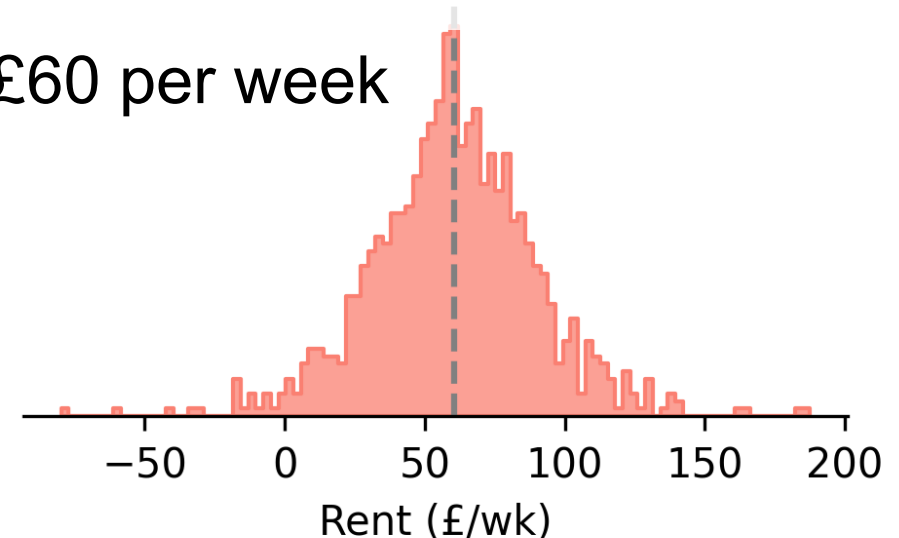
Global



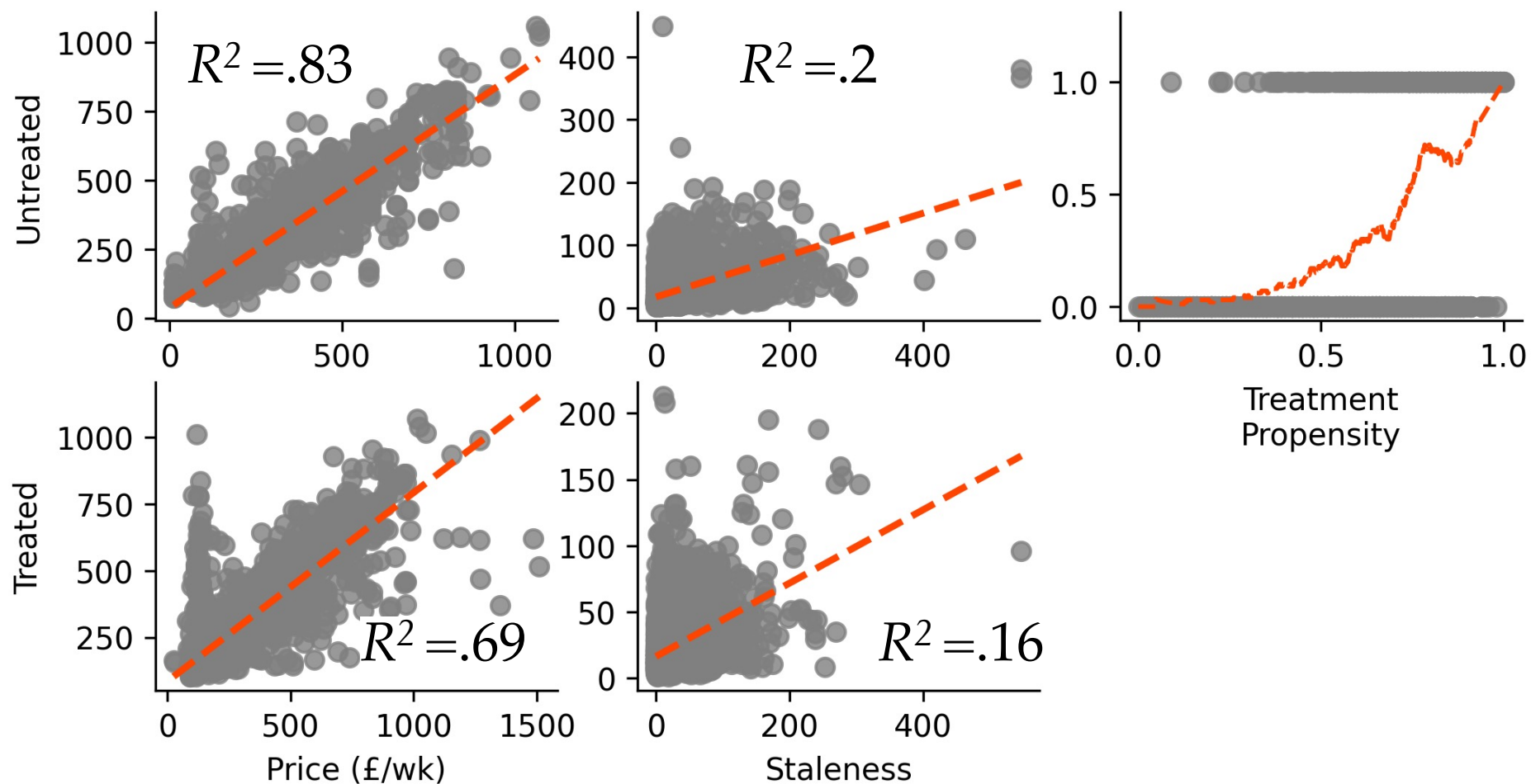
Boundary



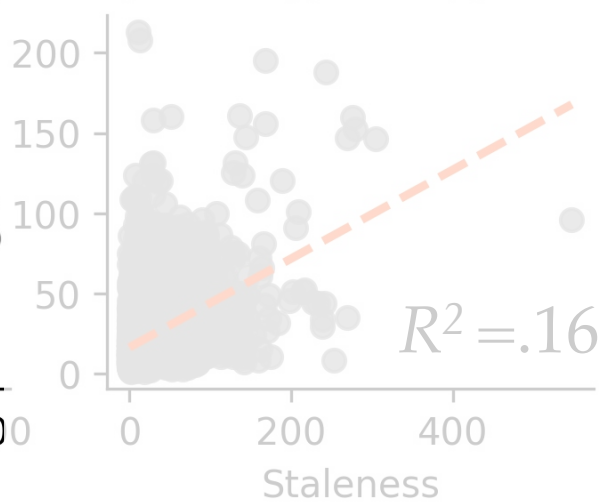
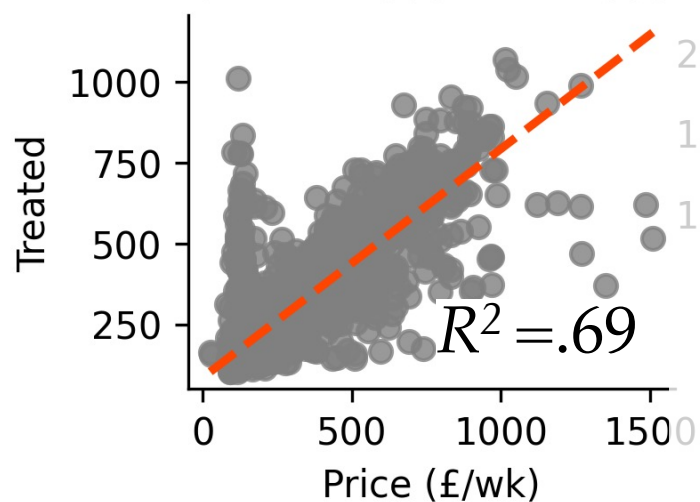
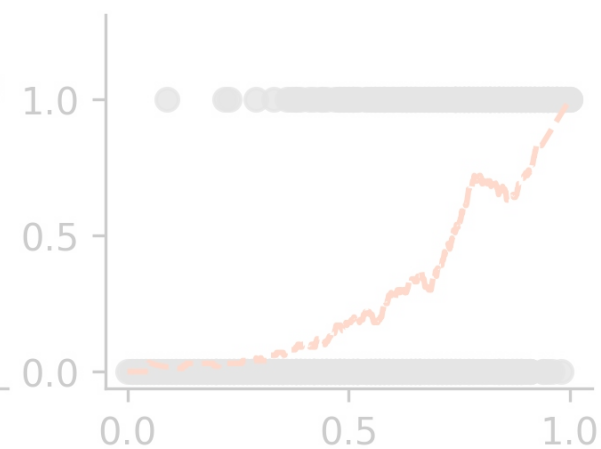
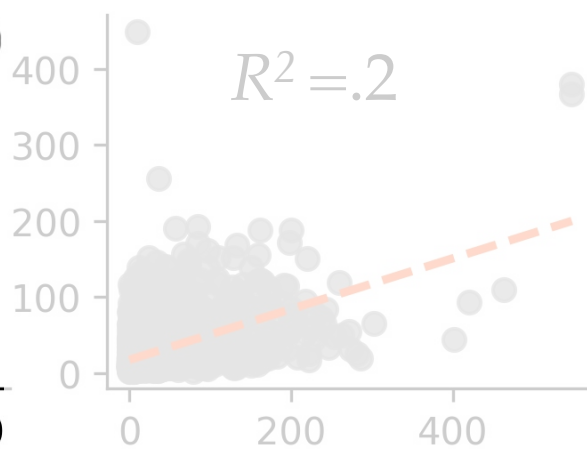
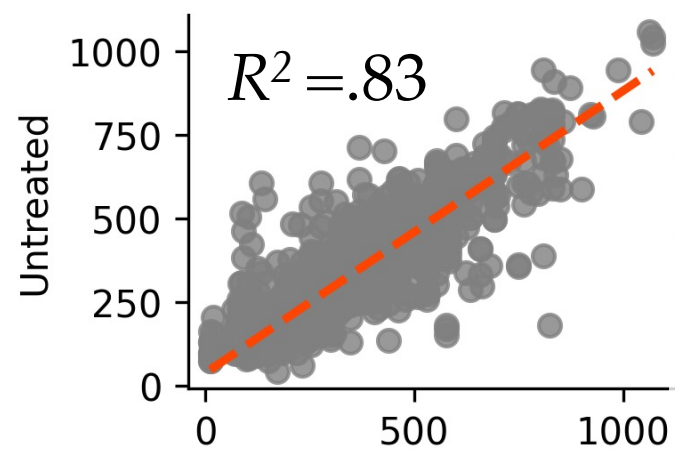
+£60 per week



SPATIAL DIFFERENCE IN DISCONTINUITY

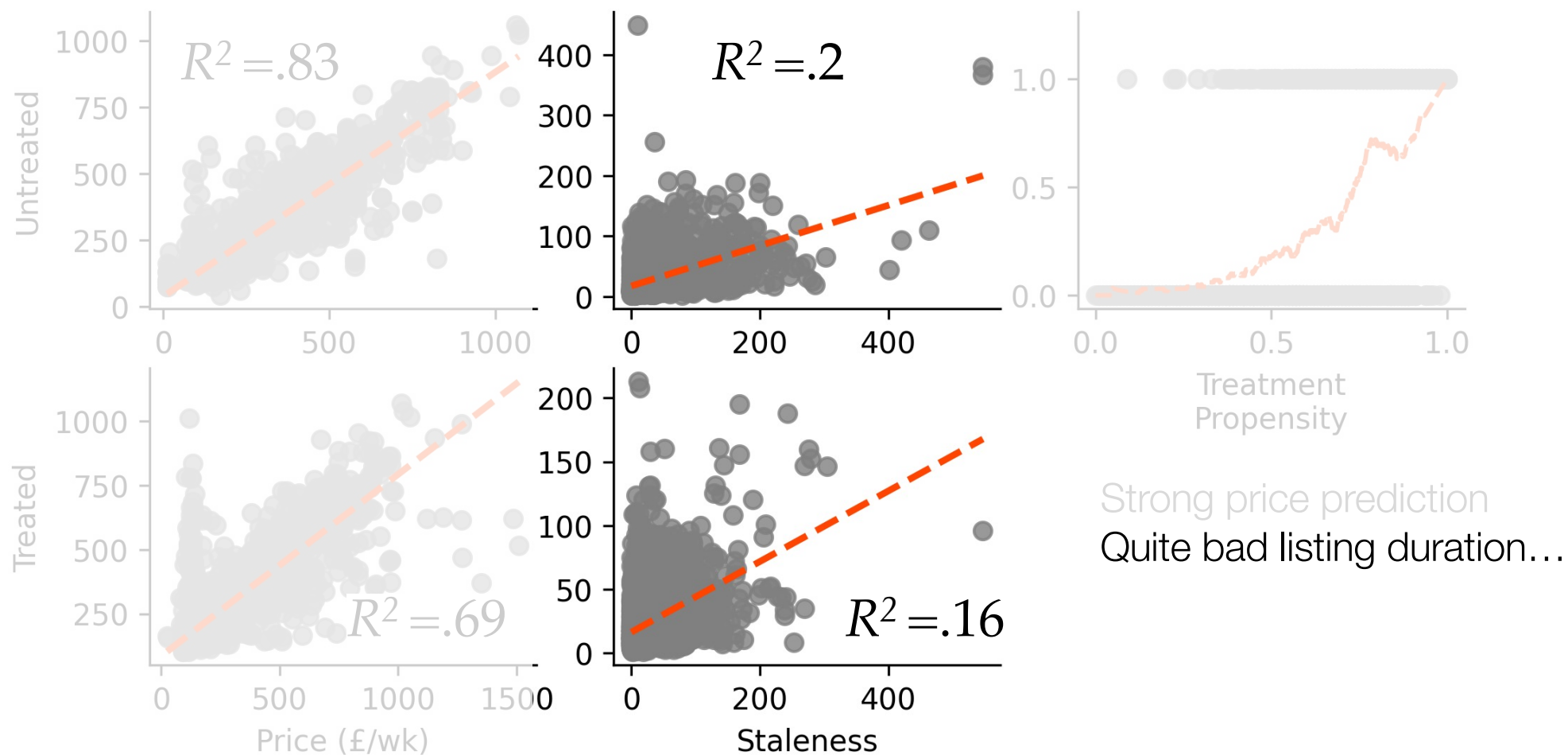


XLEARNER

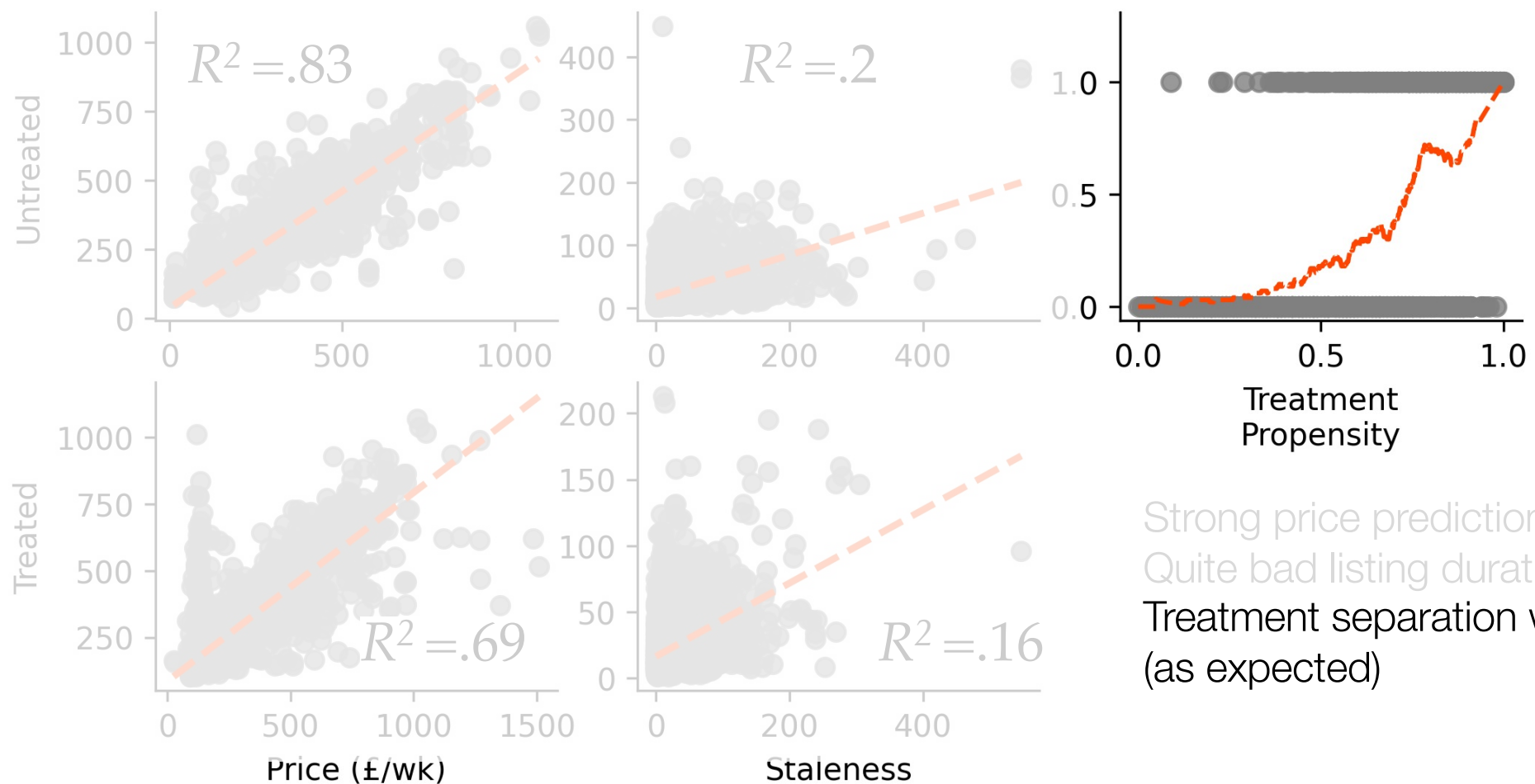


Strong price prediction

XLEARNER

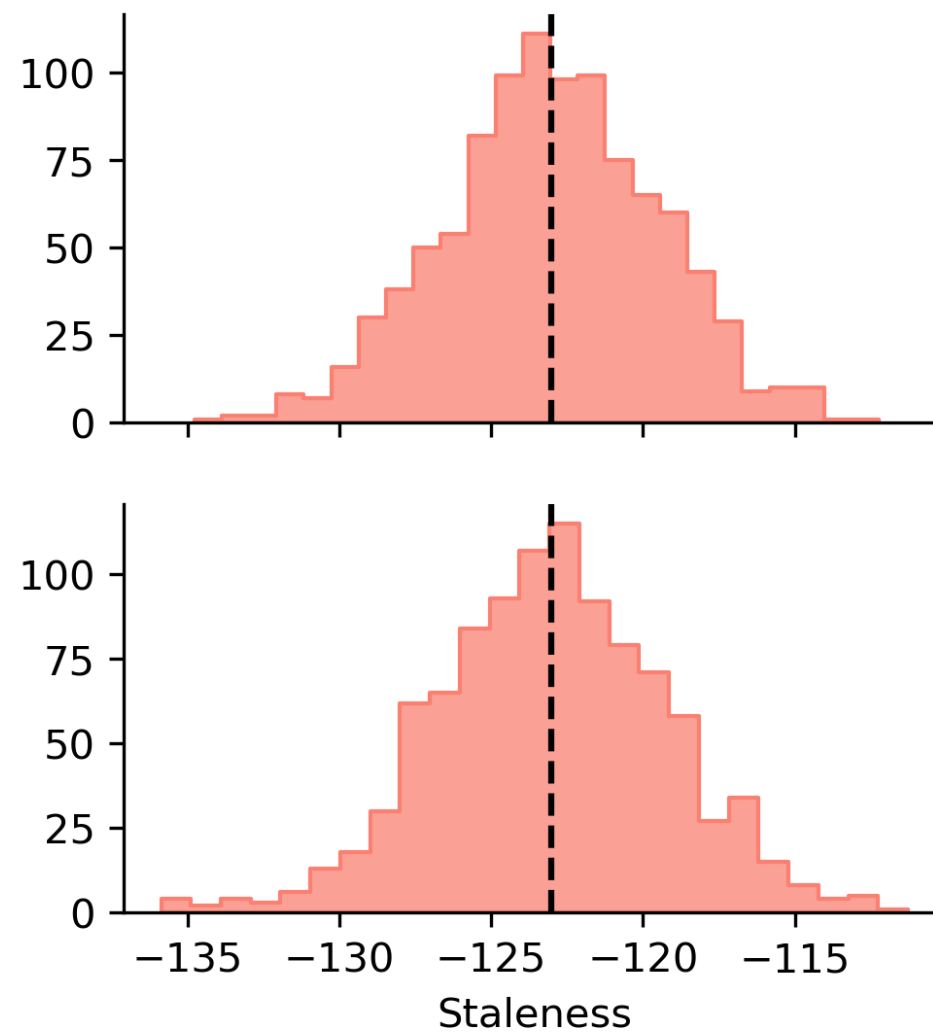
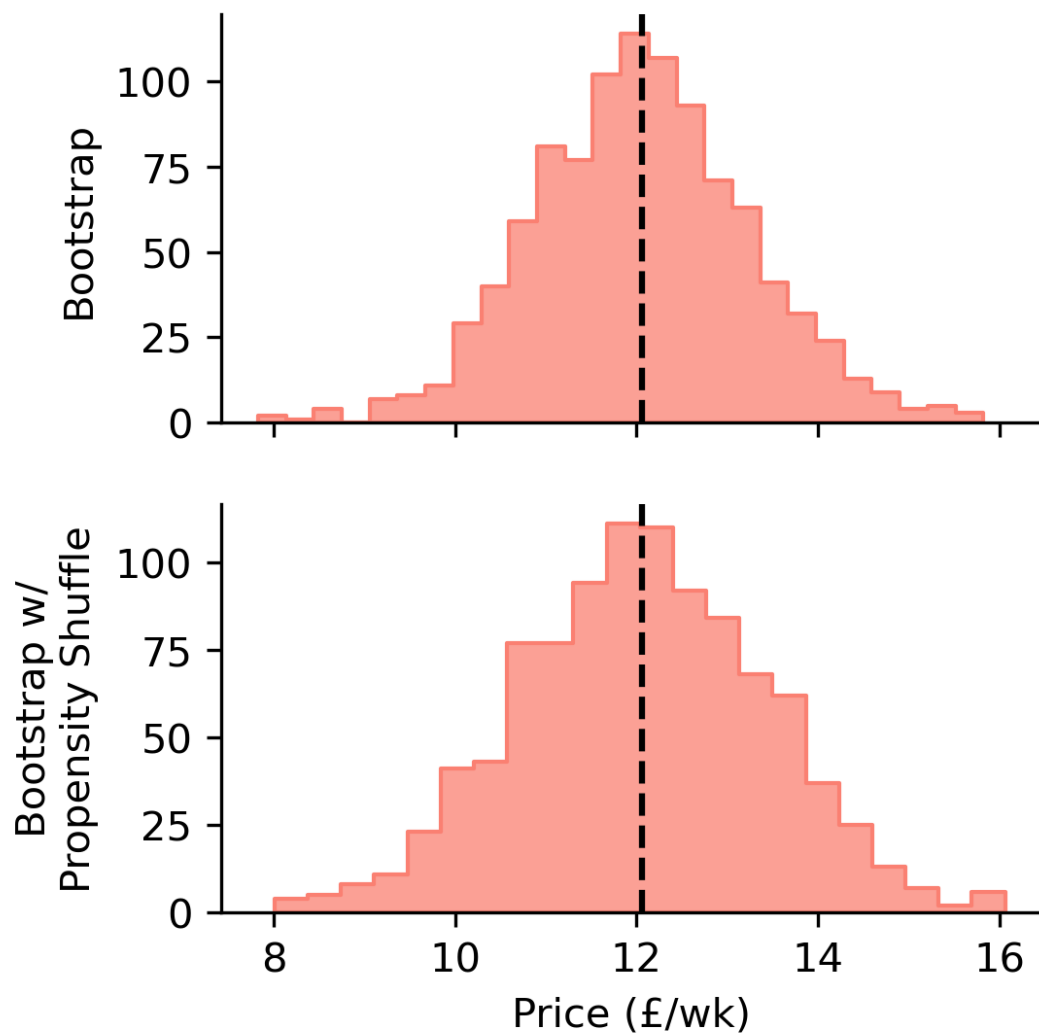


XLEARNER

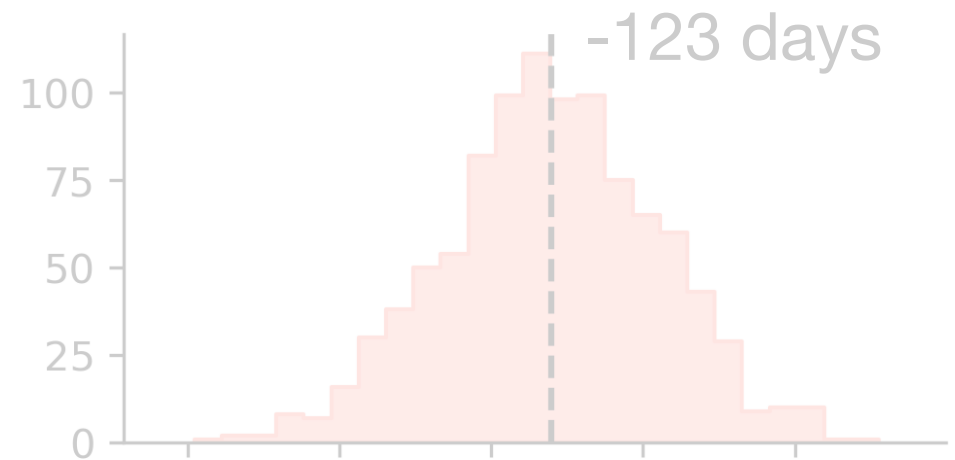
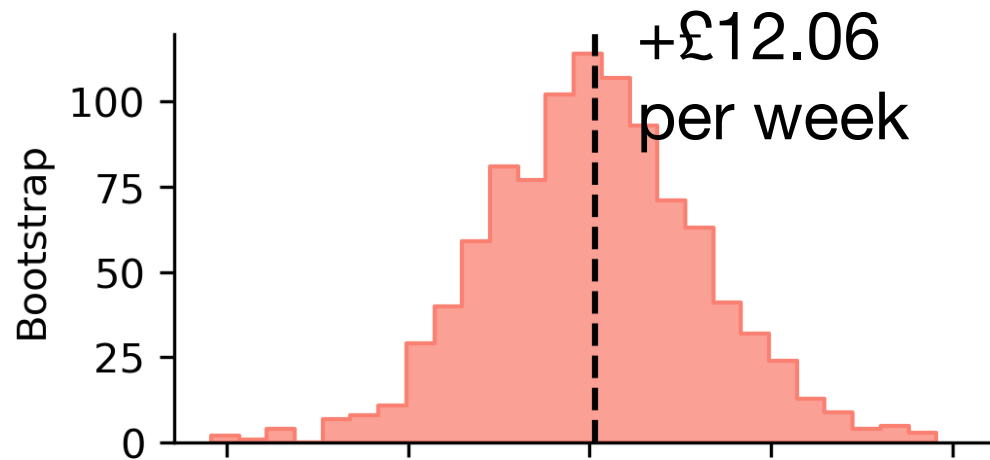


Strong price prediction
Quite bad listing duration...
Treatment separation weak
(as expected)

XLEARNER

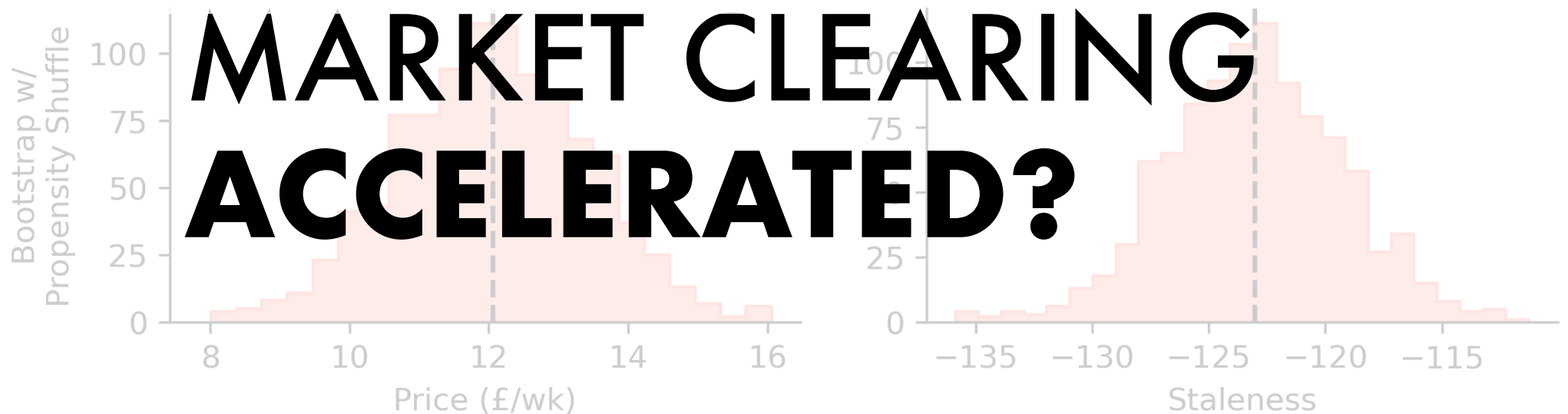
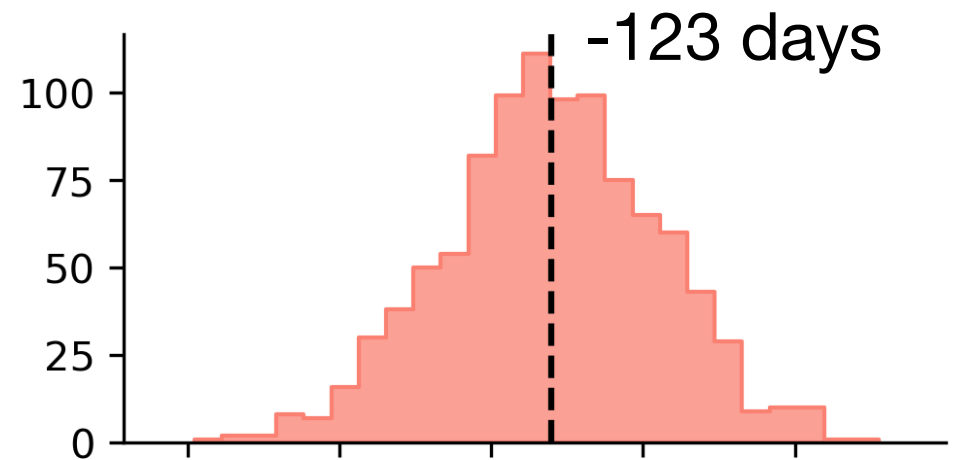
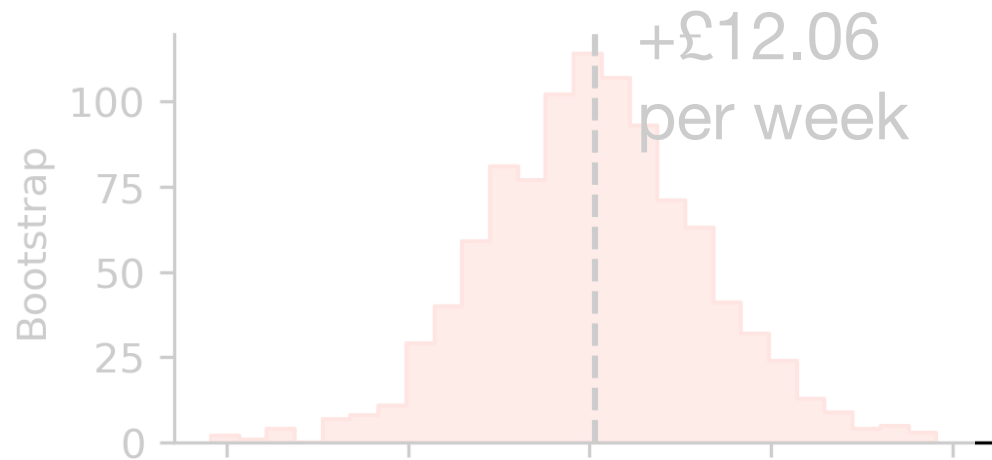


XLEARNER



**RENT GROWTH WAS
HIGHER WITHIN DLA
NEAR BOUNDARY**

XLEARNER



XLEARNER

DLA Adoption caused faster rent growth, with an uncertain effect on clearing times for private rental properties, after controlling for COVID-19, stock imbalance, and path dependence.

Landlord concerns about fewer rentals & higher rent levels seem founded.

Need access to property register to determine partial treatment, new rental market entrants

SLA OR NAY?

SLA OR NAY?



CAUSAL IMPACTS OF BRISTOL DLAS

Levi John Wolf

levi.john.wolf@bristol.ac.uk

Separate modelling into treatment models and control models. Each gets

$$Y_j \sim Y_p + X_t + e_j$$

Compute gap between observed and cfact prediction. For instance, if j is treated, calculate $\tau = Y_j - \hat{Y}_j$ with prediction from untreated model.

Estimate τ /ITE models separately again

XLEARNER DETAILS (1)

Estimate propensity to be treated p_j using

$$T_j \sim X_j + Y_j$$

Calculate propensity-weighted average treatment:

$$ATE = \Sigma (\tau_u (1 - p_j) + \tau_t p_j)$$

τ_t is the imputed ITE assuming individual j is treated, while τ_u is imputed ITE assuming not.

XLEARNER DETAILS (2)